

## CASINO BUSINESS PRESENTATION

# SPIT SHINE

The Variable-Table Game of Show, Spit, Shine & Win

Complete Deal & Play Procedures, Rules, Odds, Audit Data, and Integrity Controls



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*Prepared by the Spit Shine Development Team - Audit reference SS-AUDIT-2026-001*

# Document Control

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## Table of Contents

Document Control

2

Table of Contents

2

1. Executive Summary

6

At a glance

6

Headline audited numbers

6

Terms and how to read the values in this guide

7

Ambiguities resolved in this edition

7

How to use this guide

8

Revision history

8

2. Game Overview & Design Intent

9

Design lineage

9

3. Game Modes & Player Configurations

10

3.1 Competitive mode (default)

10

3.2 Cooperative mode - specialty/event loss-leader

10

3.3 Player counts and seat configurations

11

3.4 Mode selection rules

11

3.5 Solo vs dealer (heads-up)

11

4. Equipment & Table Layout

13

|                                                             |           |
|-------------------------------------------------------------|-----------|
| Equipment checklist . . . . .                               | 13        |
| Seating and deal order . . . . .                            | 13        |
| <b>5. Card Values &amp; Scoring</b>                         | <b>14</b> |
| Rank point values . . . . .                                 | 14        |
| Scoring examples . . . . .                                  | 14        |
| Ace duality in practice - further worked examples . . . . . | 14        |
| Exact hand-score distribution (best 2 of 4 cards) . . . . . | 15        |
| <b>6. Hand Evaluation &amp; Tie-Breaks</b>                  | <b>16</b> |
| Evaluation algorithm . . . . .                              | 16        |
| Showdown order (tie-breaks) . . . . .                       | 16        |
| Worked tie-break examples . . . . .                         | 16        |
| <b>7. How to Deal - Table Procedures</b>                    | <b>17</b> |
| 7.1 Pre-game preparation . . . . .                          | 17        |
| 7.2 Round flow - dealer script . . . . .                    | 17        |
| 7.3 Shoe and security controls . . . . .                    | 17        |
| 7.4 Worked dealing examples . . . . .                       | 18        |
| <b>8. How to Play - Phase-by-Phase Player Guide</b>         | <b>19</b> |
| Phase 1 - Private Reveal . . . . .                          | 19        |
| Phase 2 - Showing . . . . .                                 | 19        |
| Phase 3 - Bow Out . . . . .                                 | 19        |
| Phase 4 - Spit . . . . .                                    | 19        |
| Phase 5 - Shine . . . . .                                   | 20        |
| Phase 6 - Selection . . . . .                               | 20        |
| Phase 7 - Showdown & Payout . . . . .                       | 20        |
| Decision guides (reference cards) . . . . .                 | 20        |
| <b>9. Roles &amp; Responsibilities</b>                      | <b>22</b> |
| Responsibility matrix . . . . .                             | 22        |
| 9.1 The Player . . . . .                                    | 22        |
| 9.2 The Dealer . . . . .                                    | 22        |
| 9.3 The Pit Manager . . . . .                               | 23        |
| 9.4 Suspected-problem procedure . . . . .                   | 24        |
| <b>10. Dealer Procedures &amp; Payout Math</b>              | <b>25</b> |

|                                                                      |           |
|----------------------------------------------------------------------|-----------|
| 9.1 Pot accounting . . . . .                                         | 25        |
| 9.2 Payout examples . . . . .                                        | 25        |
| 9.3 Dealer's own play . . . . .                                      | 25        |
| 9.4 Disputes and corrections . . . . .                               | 25        |
| 9.5 Worked payouts with chip values . . . . .                        | 25        |
| <b>11. Limits, Bankroll &amp; Money Management</b>                   | <b>27</b> |
| 11.1 Chip denominations and ante limits . . . . .                    | 27        |
| 11.2 Player bankroll guidance . . . . .                              | 27        |
| 11.3 Cash and chip flow . . . . .                                    | 27        |
| 11.4 Responsible gaming and jurisdictional limits . . . . .          | 28        |
| <b>12. Daily Turnover &amp; Revenue Model</b>                        | <b>29</b> |
| 12.1 Definitions . . . . .                                           | 29        |
| 12.2 Model assumptions (must be validated in floor trials) . . . . . | 29        |
| 12.3 Per-round economics . . . . .                                   | 29        |
| 12.4 Daily turnover scenarios (worked) . . . . .                     | 30        |
| 12.5 Monthly and multi-table projections . . . . .                   | 30        |
| 12.6 Player session example (funds flow) . . . . .                   | 30        |
| 12.7 Calibration options for market fit . . . . .                    | 30        |
| 12.8 Round-time and pace model . . . . .                             | 31        |
| <b>13. Game State Machine &amp; Canonical Rules</b>                  | <b>33</b> |
| Round state flow . . . . .                                           | 33        |
| The table in every state . . . . .                                   | 33        |
| Value flow through the round . . . . .                               | 35        |
| Canonical rules reference (v11) . . . . .                            | 35        |
| State invariants . . . . .                                           | 35        |
| <b>14. Audit Data - Odds, EV &amp; Fairness</b>                      | <b>36</b> |
| 14.1 Exact probabilities . . . . .                                   | 36        |
| 14.2 Round outcomes - competitive mode (20,000 rounds) . . . . .     | 36        |
| 14.3 Winner score distribution . . . . .                             | 37        |
| 14.4 House edge - corrected mathematics . . . . .                    | 37        |
| 14.5 Cooperative mode economics and rebalancing . . . . .            | 38        |
| 14.6 Rule variants - what each mechanic contributes . . . . .        | 39        |
| 14.7 Seat-position fairness . . . . .                                | 39        |

14.8 Solo vs dealer (heads-up) odds . . . . . 40

14.9 Variable-table simulation (drop-in/out, mixed antes) . . . . . 40

14.10 One-million-round verification . . . . . 41

Variable-table metrics - 1,000,000 rounds . . . . . 43

14.11 Two-million-round hand audit & 24-hour distribution . . . . . 43

24-hour distribution . . . . . 46

14.12 Highly variable 2,000,000-round game day . . . . . 47

The variable day across 24 hours . . . . . 49

14.13 Player odds display & verification . . . . . 52

The published odds card (three decimal places) . . . . . 52

**15. Integrity, Fairness & Regulatory Readiness 53**

15.1 Randomness . . . . . 53

15.2 Audit trail . . . . . 53

15.3 Regulatory readiness checklist . . . . . 53

**16. Edge Cases, Official Rulings & Ambiguities 54**

16.1 Table and seating edge cases . . . . . 54

16.2 Deal and card-handling edge cases . . . . . 54

16.3 Phase-level edge cases . . . . . 54

16.4 Showdown and payout edge cases . . . . . 55

16.5 Digital and POC behavior . . . . . 55

16.6 Official rulings quick reference . . . . . 56

16.7 Open questions (decision required before certification) . . . . . 56

**17. Known Issues & Remediation 57**

**18. Appendix A - Glossary 58**

**19. Appendix B - Worked Example: One Full Round 59**

B.1 Initial deal . . . . . 59

B.2 Phase decisions . . . . . 59

B.3 Spit swaps (Phase 4) . . . . . 59

B.4 Showdown and payouts . . . . . 59

**20. Appendix C - Data Provenance & Reproduction 61**

# 1. Executive Summary

Spit Shine is a proprietary casino table game for one to four players against the dealer, played with a 104-card shoe (two standard 52-card decks). Each seat receives four cards and, across seven clearly defined phases - Private Reveal, Showing, Bow Out, Spit, Shine, Selection, and Showdown - builds the highest-scoring two-card hand. The game combines the bluffing depth of poker, the fixed-dealer structure of blackjack, and the clean point-comparison of baccarat, but its signature mechanics (Spit card exchange and Shine redraw) create a level of player agency absent from most table games.

The table is not fixed at any player count. It seats any number of players from one to four against the dealer, and players may join or drop out freely between rounds: a player who joins the next round posts an ante and receives a fresh hand, a player who leaves simply does not post for the next round, and empty seats post no ante and take no cards. The pot, rake, and payouts scale with the occupied seats, and every seat chooses its own ante within the published 1-500 chip range.

This guide is the complete operating manual for the game: how to deal it, how to play it, every rule and resolution, the full statistical audit from every perspective (player, dealer, house, regulator), and the integrity controls required for a regulated casino environment. Every figure in this document is reproducible from the canonical engine and the audited simulation dataset.

## At a glance

| Attribute      | Value                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------|
| Game           | Point-comparison table game with seven decision phases                                               |
| Players        | 1-4 player seats vs the dealer (heads-up to full table); players drop in/out between rounds          |
| Equipment      | 2 standard 52-card decks (104 cards), fresh shoe per round, cut card recommended                     |
| Ante           | Variable per seat, 1-500 chips (denominations 1, 5, 25, 100, 500)                                    |
| House edge     | 25% pot rake; audited hold 24.462% (competitive)                                                     |
| Modes          | Competitive PVP (standard, dealer plays) and Cooperative specialty/event mode (optional loss-leader) |
| Round duration | 1.962-3.407 minutes by skill mix (17.603-30.536 rounds/hour, Section 12.8)                           |
| Reference      | Canonical engine v11; audit SS-AUDIT-2026-001; 20,000-round seeded Monte Carlo per scenario          |

## Headline audited numbers

| Scenario                                        | Key figure                       | Detail                                                                                       |
|-------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------|
| Full table (4 players + dealer, 100 ante)       | Dealer wins 24.610%; YOU 17.830% | YOU EV -33.875/round; house 122.5/round                                                      |
| Solo heads-up (competitive, 100 ante)           | YOU wins 47.975%                 | YOU EV -26.815/round; house 49.4/round (24.7% hold)                                          |
| Variable table (drop-in/out, mixed 1-500 antes) | Average 3.59 players             | Handle 565 chips/round; house 138.4/round; hold 24.479%                                      |
| Cooperative (specialty/event)                   | Team wins 78.800%                | Player EV +3.103/round; house cost -12.414/round (intentional marketing loss - Section 14.5) |

| Scenario                     | Key figure                                                      | Detail                                                             |
|------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|
| Double Aces                  | 2.895% per hand; 13.917% dealt per round                        | A+A = 1000 points; ~17% of rounds finish with A+A after Spit/Shine |
| Exact-tie rounds             | 3.165% of rounds                                                | Pot splits among tied winners; rounding rake to house              |
| One-million-round audit      | Competitive: YOU 17.73%, dealer 24.66%                          | Section 14.10 - 3 x 1M rounds with 95% CI and charts               |
| Two-million-round hand audit | 10M hands tracked; hold 24.51%; 24h model \$70.6k/day house win | Section 14.11 - per-round hand/point tracking + 24h charts         |
| Round-time model             | 17.6-30.5 rounds/hour by player/dealer skill                    | Section 12.8 - includes explanation and issue-fix time             |
| Highly variable 2M day       | Average 3.006 players; hold 24.462%                             | Section 14.12 - uniform 1-500 antes, 800k drop-ins/outs            |

## Terms and how to read the values in this guide

| Term                     | Meaning                                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| EV (expected value)      | The average chips a seat wins or loses per round over many rounds. A negative EV means the seat loses on average; a positive EV means it gains. |
| Hold / hold percentage   | House win divided by handle - the share of every chip wagered that the house keeps.                                                             |
| Handle                   | Total chips wagered (antes posted) in a period.                                                                                                 |
| Rake                     | The house's 25% cut of the pot at showdown.                                                                                                     |
| Ante                     | The wager each seat posts before cards are dealt (1-500 chips).                                                                                 |
| Spread                   | The ratio between the largest and smallest ante at one table (5x recommended).                                                                  |
| CI (confidence interval) | The range around a simulated value that contains the true value with 95% probability.                                                           |
| pp (percentage points)   | The arithmetic difference between two percentages, e.g., 24.657% vs 20.000% is 4.657 pp.                                                        |
| rph                      | Rounds per hour.                                                                                                                                |
| 1v1 / heads-up           | One player versus the dealer.                                                                                                                   |
| P(win)                   | Probability of winning, shown as a percentage.                                                                                                  |
| RNG                      | Random number generator - the source of shuffle randomness.                                                                                     |
| GLI                      | Gaming Laboratories International - an independent game-certification laboratory.                                                               |
| AML / KYC                | Anti-money-laundering and know-your-customer compliance controls.                                                                               |

Every value in this guide is shown to three decimal places unless it is a discrete count (rounds, hands, or chips issued). All monetary values are in chips; multiply by the chip value to get currency. Percentages are exact to the displayed precision and are reproducible from the audited simulation datasets.

## Ambiguities resolved in this edition

- Exact ties split the pot (score -> rank -> suit -> split); any integer remainder is retained by the house as rounding rake and logged.

- The Spit-received card is private to the spitter; a target can receive at most one Spit per round; the dealer never Spits or Bows Out.
- Tables are variable: 1-4 player seats, per-seat antes from 1-500 chips, players drop in/out between rounds, empty seats post no ante.
- Mixed antes transfer EV toward small bettors, so a 5x ante spread limit is recommended (Section 14.9).
- Cooperative mode's current economics are house-favorable; a rebalance package (dealer 1-card Shine, dealer ante, 125-chip team payout) brings player EV to about -1.8 chips/round.
- Every remaining edge case and official ruling is collected in Section 15.

## How to use this guide

| Reader                           | Start here                                                             |
|----------------------------------|------------------------------------------------------------------------|
| Dealers / table supervisors      | Sections 7-10: procedures, phase-by-phase duties, payout math, rulings |
| Players                          | Section 8 and Appendix B: phase guide and a complete worked round      |
| Operators / business development | Sections 3, 11-13: modes, limits, turnover model, canonical rules      |
| Regulators / certification labs  | Sections 14-16: audit data, integrity controls, edge-case rulings      |

## Revision history

| Version | Date           | Changes                                                                                                                                                                                                                                                                                  |
|---------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | August 6, 2026 | Initial 24-page presentation guide                                                                                                                                                                                                                                                       |
| 1.1     | August 6, 2026 | Audit finalized (SS-AUDIT-2026-001); companion references                                                                                                                                                                                                                                |
| 1.2     | August 6, 2026 | Variable tables: 1-4 players, drop-in/out, 1-500 chip antes, heads-up solo (36 pages)                                                                                                                                                                                                    |
| 1.3     | August 6, 2026 | Expanded executive summary, edge-case rulings (Section 16), decision flowcharts, phase-state depictions, and additional charts and examples                                                                                                                                              |
| 1.4     | August 6, 2026 | Dedicated roles section (Player, Dealer, Pit Manager) with integrity watchlists; executive summary states the flexible 1-4 player table and drop-in/out rules explicitly                                                                                                                 |
| 1.5     | August 6, 2026 | One-million-round verification: 3 x 1,000,000 rounds (competitive, variable table, solo) with 95% confidence intervals and new metric charts (Section 14.10)                                                                                                                             |
| 1.6     | August 6, 2026 | Two-million-round hand audit: every seat's hand and point totals tracked (10M hands), aggregates and a 5,000-round raw sample, plus a 24-hour distribution model (Section 14.11)                                                                                                         |
| 1.7     | August 6, 2026 | Clarity pass (EV and all terms defined, three-decimal values), round-time model for beginner/intermediate/advanced players and dealers, highly variable 2M game day, cooperative loss-leader repositioning with conversion economics, and the player odds display & verification section |

The audit's headline finding is that the game's published odds were materially incorrect (the Double-Aces frequency was understated by roughly 30x and the house edge by 5x). This guide presents the corrected, simulation-verified numbers and the canonical rule set that every future implementation must follow.



## 2. Game Overview & Design Intent

Spit Shine is designed to occupy the space between pure-chance games (baccarat) and skill-heavy games (poker): simple enough for a casual player to learn in one round, deep enough for regulars to develop a real strategy. Every round contains exactly seven decision phases, and every phase creates observable information - shown cards, spit targets, shine behavior - that skilled players can exploit.

### Design lineage

| Feature            | Poker      | Blackjack | Baccarat   | Spit Shine                  |
|--------------------|------------|-----------|------------|-----------------------------|
| Decks              | 1          | 6-8       | 8          | 2                           |
| Players vs         | Each other | Dealer    | Banker     | All seats                   |
| Hand construction  | 5 cards    | 2+ cards  | 2-3 cards  | 4 cards to 2                |
| Decision points    | 4          | 1         | 0          | 7 phases                    |
| Player information | Partial    | 1 up card | None       | Partial + Spit intel        |
| Bluffing           | Core       | None      | None       | Show/hide + Bow Out         |
| Card exchange      | Draw poker | None      | None       | Spit phase                  |
| Discard / redraw   | None       | None      | None       | Shine phase                 |
| Safety valve       | Fold       | Surrender | None       | Bow Out (25% refund)        |
| House edge source  | Rake       | Rules     | Commission | 25% pot rake + dealer Shine |

### Unique innovations

- Spit: exchange one face-down card with an opponent. You see what you receive; they do not see what you gave. Each seat can be spat at once per round.
- Shine: optional redraw. Players discard 0-1 face-down cards; the dealer may discard 0-2 - the house's structural edge.
- Bow Out: hidden hands may fold and recover 25% of the ante, a risk-management lever that rewards discipline.
- Showing: revealing cards is a commitment (it locks out Bow Out) and an information weapon.

### 3. Game Modes & Player Configurations

Spit Shine supports two officially specified modes. Both use the same table geometry (up to four player seats plus the dealer) and the same seven-phase round, but the payout structures and the dealer's role differ. This section is the authoritative reference for how many players a table can seat, how empty seats are handled, and what each mode pays.

#### 3.1 Competitive mode (default)

Competitive mode seats four players and the dealer at a full table; all seats ante within the 1-500 chip limits. The dealer is a full participant - the dealer can win the pot like any seat - but is barred from Bow Out and Spit, and may discard two cards in Shine instead of one. The highest-scoring hand at showdown wins 75% of the pot after the 25% rake.

| Item                | Specification                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------|
| Seats               | 1-4 players + dealer (variable; full table = 5 seats)                                                      |
| Ante                | Variable 1-500 chips per seat, including the dealer (examples below at 100)                                |
| Pot (full table)    | 500 chips                                                                                                  |
| Winner receives     | 375 chips (pot - 25% rake of 125)                                                                          |
| Dealer restrictions | No Bow Out, no Spit; may Shine 0-2 cards                                                                   |
| Example             | YOU wins with K+Q (350). Pot 500, rake 125, payout 375. YOU nets +275 chips for the round (100-chip ante). |

#### 3.2 Cooperative mode - specialty/event loss-leader

Cooperative mode is NOT a standard table offering. It is a specialty game derived from the original Spit Shine rules, used only as an attraction device: casinos run it at events and promotions, accept a net potential monetary loss, and convert the players who enjoyed it to the profitable competitive PVP mode. Standard tables do not have to include it; this guide documents it so its economics are explicit and controlled.

##### The original cooperative rules (reference only)

| Item                  | Specification                                                                |
|-----------------------|------------------------------------------------------------------------------|
| Seats                 | 1-4 players + dealer (variable)                                              |
| Ante                  | Variable 1-500 chips per player (examples at 100); dealer antes 0            |
| Pot (full table)      | 400 chips                                                                    |
| Team win              | All non-bowed players split 300 chips (75 each)                              |
| Dealer win            | House keeps the full 400 chips                                               |
| Audited team win rate | 70.275% (20,000 rounds)                                                      |
| Audited player EV     | -46.138 chips per round (house edge ~46.138%) - NOT recommended as published |

##### The recommended attraction package (audited)

| Attraction rule              | Value                                                                          |
|------------------------------|--------------------------------------------------------------------------------|
| Dealer Shine                 | 1 card (same as players)                                                       |
| Dealer ante                  | 100 chips (same as players)                                                    |
| Team win payout              | 140 chips gross per active player (house subsidizes 20 chips on each team win) |
| Audited team win rate        | 78.800% (100,000 rounds)                                                       |
| Player expected value (EV)   | +3.103 chips per round                                                         |
| House cost per round         | -12.414 chips (intentional marketing loss)                                     |
| House take per round (gross) | +87.586 chips before the dealer ante is counted                                |

This package is deliberately player-positive by about 3.103 chips per round - enticing enough to fill an event table, but not exploitable: the house still wins 21.212% of rounds outright and the player edge is comparable to a modest promotion. The full economics and the conversion math that justify the loss are in Section 14.5.

### 3.3 Player counts and seat configurations

| Configuration       | Seats                     | Use case                         | Notes                                                                         |
|---------------------|---------------------------|----------------------------------|-------------------------------------------------------------------------------|
| Heads-up (solo)     | 1 player + dealer         | Single player vs dealer          | Fully supported in both modes; competitive heads-up recommended (Section 3.5) |
| Live short-handed   | 2-3 players + dealer      | Low-traffic hours                | Drop in/out between rounds; empty seats do not ante                           |
| Live standard table | 4 players + dealer        | Recommended casino configuration | Players join/leave between rounds; capacity capped at 4                       |
| Digital POC / demo  | 1 human + 0-3 AI + dealer | Sales demo, training, simulation | AI fill optional; identical rules to live play                                |

Tables are intentionally variable: players may drop in or out between rounds, and every seat may choose its own ante within the published 1-500 chip range. Seats never change mid-round; empty seats post no ante and take no cards. Every seat plays an independent hand, so the rules and odds per seat are identical at any table size - a short-handed table simply has a smaller pot. Casinos should publish the table minimums, cap the table at four player seats, and keep empty seats clearly marked.

### 3.4 Mode selection rules

- Mode is announced before the first ante of a session and cannot change mid-shoe.
- A table may run different modes on different shifts, but a single table cannot mix modes in one round.
- Side wagers are not part of the current rules; any future side bet requires independent math certification before deployment.

### 3.5 Solo vs dealer (heads-up)

A single player can play the dealer heads-up in both modes. The canonical engine and the Triple-A client support 1-4 player seats, and the heads-up table is the natural low-traffic configuration. The player's win rate is far higher than at a full table - they only need to beat the dealer - but the 25% rake still makes the game house-positive.

| Metric (100-chip ante, 20k rounds) | Competitive heads-up           | Cooperative heads-up                          |
|------------------------------------|--------------------------------|-----------------------------------------------|
| Player win rate                    | 48.0%                          | 43.5%                                         |
| Dealer win rate                    | 53.4%                          | 56.5%                                         |
| Player EV per round                | -26.8 chips                    | -65.2 chips                                   |
| House win per round                | +49.4 chips (24.7% hold)       | +65.2 chips (65.2% hold)                      |
| Pot                                | 200 chips (both ante)          | 100 chips (dealer antes 0)                    |
| Verdict                            | Recommended solo configuration | House-favorable; not recommended as published |

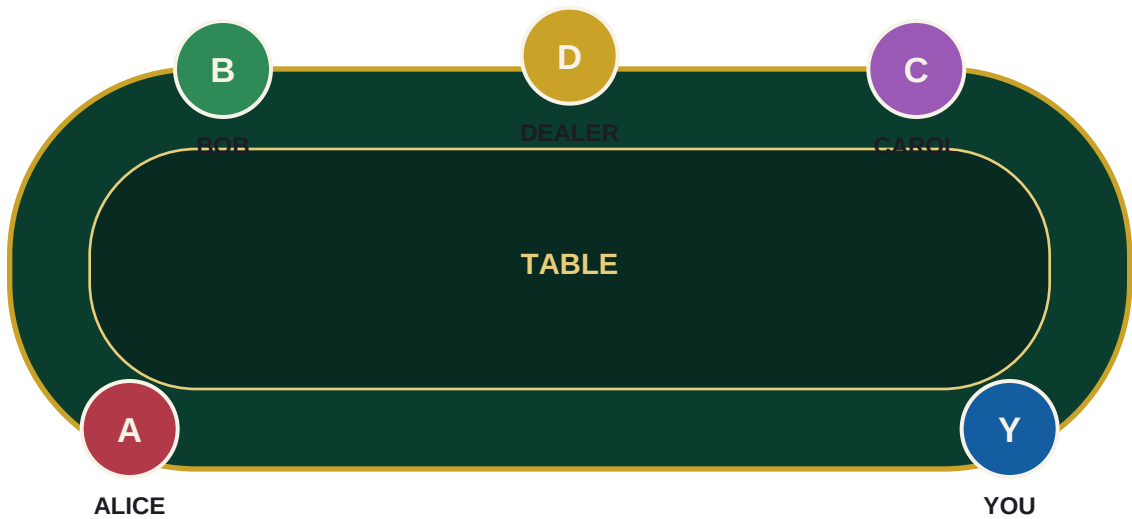
Recommended solo rule: competitive heads-up at a variable ante within table limits. The cooperative heads-up payout (player wins 75 chips against a 100-chip ante; dealer win forfeits the whole pot) gives the house a 65.2% hold and should be rebalanced or withdrawn, consistent with Section 14.5.

# 4. Equipment & Table Layout

## Equipment checklist

| Item         | Specification                               | Control requirement                                             |
|--------------|---------------------------------------------|-----------------------------------------------------------------|
| Cards        | Two unopened 52-card decks, identical backs | Inspect for marks; count 52 per deck before and after play      |
| Shoe         | Dealing shoe for up to 104 cards            | Tamper-evident; fixed position under surveillance               |
| Cut card     | Standard plastic divider                    | Insert 13-15 cards from the bottom (approx. 85-87% penetration) |
| Chips        | Standard 25/100/500 denominations           | House drop box for rake; chip fills recorded                    |
| Table        | 5 seats + dealer, 8-12 ft oval              | Dealer station with shoe, drop box, discard rack                |
| Surveillance | Overhead + dealer camera                    | Cards, shoe, and drop box visible at all times                  |

## Seating and deal order



*Deal order (canonical): YOU, ALICE, BOB, CAROL, DEALER - one card at a time*

The canonical engine deals round-robin one card at a time in seat order: YOU, ALICE, BOB, CAROL, DEALER. A floor operation may adopt standard clockwise dealing starting left of the dealer and dealing the dealer last; the two orders are statistically equivalent. The Spit turn order follows the same fixed rotation in the reference implementation and should be rotated per round on a live floor (Section 14.7).

## 5. Card Values & Scoring

### Rank point values

| Rank | Points | Rank | Points | Rank | Points             |
|------|--------|------|--------|------|--------------------|
| 2    | 20     | 7    | 70     | Q    | 150                |
| 3    | 30     | 8    | 80     | K    | 200                |
| 4    | 40     | 9    | 90     | A    | 100 or 0 (duality) |
| 5    | 50     | 10   | 100    | A+A  | 1000 (Double Aces) |
| 6    | 60     | J    | 100    |      |                    |

### Ace duality

An Ace counts as 100 points or as 0 points, whichever maximizes the final hand score.  $A+K = 300$ ;  $A+2 = 120$ ; A alone paired with nothing is never forced to count 0 unless it is the better option. Two Aces together are the special Double Aces hand: 1000 points, the highest possible score. The maximum non-Ace pair is  $K+K = 400$ , which means no non-Ace hand can reach the 500+ range.

### Scoring examples

| Hand   | Calculation                         | Score |
|--------|-------------------------------------|-------|
| K + Q  | $200 + 150$                         | 350   |
| K + A  | $200 + 100$ (Ace = 100)             | 300   |
| 10 + J | $100 + 100$                         | 200   |
| A + 2  | $100 + 20$ (Ace = 100 beats 0 + 20) | 120   |
| 7 + 5  | $70 + 50$                           | 120   |
| A + A  | Special rule                        | 1000  |

### Ace duality in practice - further worked examples

| Hand     | Pairings considered                                                   | Best score | Why                                     |
|----------|-----------------------------------------------------------------------|------------|-----------------------------------------|
| A K 7 2  | $A+K=300$ , $A+7=170$ , $A+2=120$ , $K+7=270$ , $K+2=220$ , $7+2=90$  | 300        | Ace counts 100 with the King            |
| A 2 3 4  | $A+4=140$ , $A+3=130$ , $A+2=120$ , $4+3=70$ , $4+2=60$ , $3+2=50$    | 140        | Ace as 100 dominates every non-Ace pair |
| A 9 8 2  | $A+9=190$ , $A+8=180$ , $A+2=120$ , $9+8=170$ , $9+2=110$ , $8+2=100$ | 190        | Ace + 9 beats 9 + 8 by 20 points        |
| K K 5 3  | $K+K=400$ , $K+5=250$ , $K+3=230$ , $5+3=80$                          | 400        | The maximum non-Ace score               |
| Q Q 10 J | $Q+Q=300$ , $Q+J=250$ , $Q+10=250$ , $10+J=200$                       | 300        | Pair of Queens beats 10+J by 100        |

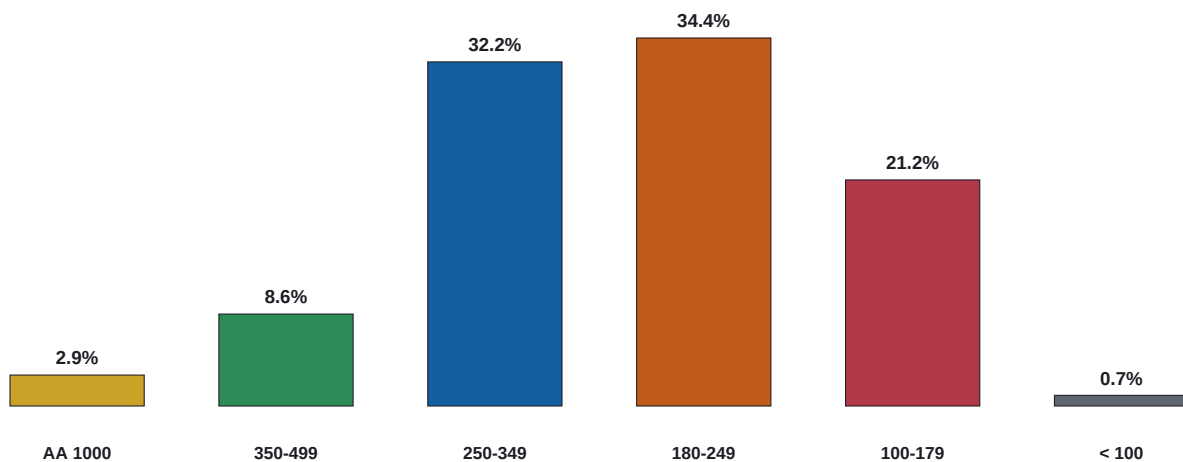
| Hand    | Pairings considered                          | Best score | Why                                      |
|---------|----------------------------------------------|------------|------------------------------------------|
| A A 2 3 | A+A=1000 (special), A+2=120, A+3=130, 2+3=50 | 1000       | Double Aces overrides all other pairings |

## Exact hand-score distribution (best 2 of 4 cards)

| Tier                         | Exact count (of 4,598,126) | Probability | Note                                             |
|------------------------------|----------------------------|-------------|--------------------------------------------------|
| Monster - Double Aces (1000) | 133,126                    | 2.895%      |                                                  |
| 500-999                      | 0                          | 0.000%      | Impossible by rule - no non-Ace pair reaches 500 |
| Strong (350-499)             | 395,702                    | 8.606%      |                                                  |
| Solid (250-349)              | 1,480,934                  | 32.207%     |                                                  |
| Marginal (180-249)           | 1,583,762                  | 34.444%     |                                                  |
| Weak (100-179)               | 972,856                    | 21.158%     |                                                  |
| Dead (< 100)                 | 31,746                     | 0.690%      |                                                  |

Source: exact enumeration of all  $C(104,4) = 4,598,126$  four-card hands in the canonical engine. Suits affect only tie-breaks, so this distribution is exact for scores.

## Exact hand-score distribution - best 2 of 4 cards



## 6. Hand Evaluation & Tie-Breaks

### Evaluation algorithm

- Step 1: If the hand contains two Aces, the score is 1000 (Double Aces).
- Step 2: For every other hand, score each Ace as 100 or 0 to maximize the total.
- Step 3: The score is the sum of the two cards' points under that rule.
- Step 4: From four cards, evaluate all six possible pairings and keep the best.

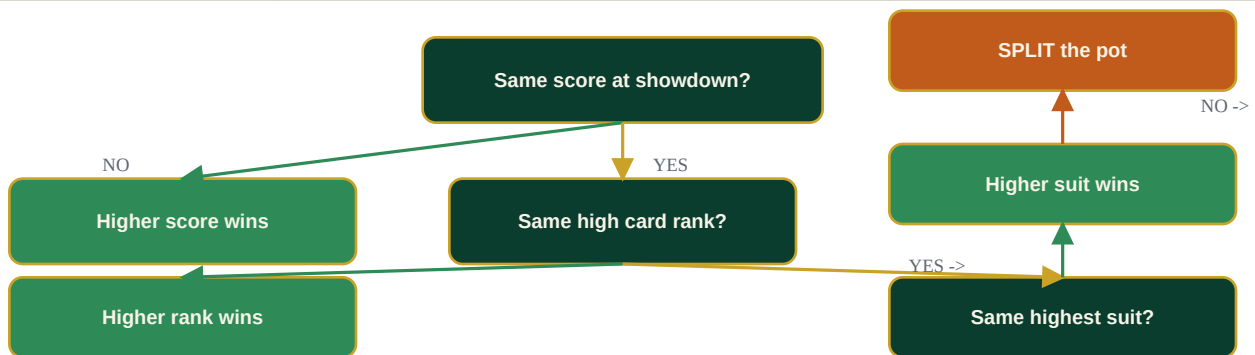
### Showdown order (tie-breaks)

When two or more seats finish with the same score, the winner is resolved in this exact order: (1) higher score; (2) higher individual card rank; (3) higher suit value (clubs 1, diamonds 2, hearts 3, spades 4). If all three are identical, the pot is split equally among the tied seats.

Exact ties are not rare: 633 of 20,000 simulated rounds (3.165%) ended in a pot split. Any table procedure must define split payouts in advance (see Section 10.4).

### Worked tie-break examples

| Scenario                           | Hands                                                               | Resolution                                                                                                    |
|------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Score tie, rank breaks it          | ALICE: A+9 = 190 vs CAROL: 10+9 = 190                               | ALICE wins - high rank Ace (14) beats 10 (11)                                                                 |
| Score and rank tie, suit breaks it | YOU: K(spades)+Q(clubs) = 350 vs ALICE: K(hearts)+Q(diamonds) = 350 | YOU wins - highest suit value spades (4) beats hearts (3)                                                     |
| Full tie - pot splits              | YOU: K(spades)+Q(hearts) = 350 vs ALICE: K(hearts)+Q(spades) = 350  | Split - both score 350, rank 13, suit-max 4. Payout 375 becomes 187 + 187 + 1 chip rounding rake to the house |



Audit note: earlier prototypes silently awarded the entire pot to the earliest seat on an exact tie. The canonical rule set adopted for casino operation splits the pot, matching the design documents.



## 7. How to Deal - Table Procedures

### 7.1 Pre-game preparation

- Open two new decks in front of surveillance; inspect every card for marks, bends, or irregularities; count 52 cards per deck.
- Shuffle: minimum seven riffle shuffles with alternating cuts, followed by a strip and a final riffle. Square the deck completely.
- Insert the cut card 13-15 cards from the bottom (recommended penetration 85-87% for a two-deck game).
- Load the shoe face-down, verify the deck's edge is even, and place the shoe in the marked position.
- Burn procedure: before the first round of a new shoe, draw one card face-down to the discard rack; it plays no part in the round.
- Announce table limits and confirm each seat's ante before dealing.

### 7.2 Round flow - dealer script

| Step | Action                                                                                                                                                       | Dealer control                                                                             |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1    | Collect antes: each active seat posts its declared ante within the published 1-500 chip limits (dealer posts an ante in competitive mode, 0 in cooperative). | Verify each ante against the table limits; place chips in the pot area; confirm the total. |
| 2    | Deal four cards to each seat, one card at a time, in seat order; dealer takes the last card of each pass.                                                    | No card may be exposed; announce 'Cards out' when complete.                                |
| 3    | Phase 1: allow private review. No table action.                                                                                                              | Do not touch cards.                                                                        |
| 4    | Phase 2: seats may show 0-4 cards. Announce each shown card as it is revealed; remind players that showing locks out Bow Out.                                | Record shown cards on the audit pad.                                                       |
| 5    | Phase 3: hidden seats may Bow Out. Pay 25% of ante (25 chips) and muck their cards.                                                                          | Remove the refund from the pot; verify pot total.                                          |
| 6    | Phase 4: each active non-dealer seat in turn may Spit once. Execute the swap without exposing the given card; reveal the received card to the spitter only.  | Mark the target seat as 'spat' - one Spit per target per round.                            |
| 7    | Phase 5: seats may Shine (discard 0-1 face-down and draw). The dealer Shines last with 0-2 discards.                                                         | Draw replacements from the shoe; discard the old cards.                                    |
| 8    | Phase 6: each seat selects its best two-card hand; the dealer selects their own pair.                                                                        | Confirm no seat has more or fewer than four cards.                                         |
| 9    | Phase 7: dealer reveals their own pair first, then each seat's pair; announce scores and determine the winner using score, rank, then suit.                  | On an exact tie, split the pot per Section 10.4.                                           |
| 10   | Payout: rake 25% of the pot to the drop box; pay the winner(s); collect and count the discard rack.                                                          | When the cut card appears, finish the round and reshuffle.                                 |

### 7.3 Shoe and security controls

- The cut card ends the shoe: stop dealing after the round in which it appears, then reshuffle using the same procedure.
- The discard rack must be visible to surveillance at all times; discards are never returned to the shoe mid-shoe.
- A dealer change or table opening requires a full shuffle and a new burn card.

- Any exposed card during the deal is treated as a dead card: remove it to the discard rack and redeal that seat's card from the shoe.

## 7.4 Worked dealing examples

| Example               | Detail                                                                                                                      | Result                                                                                                                                                      |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shoe penetration      | Cut card inserted 14 cards from the bottom of a 104-card shoe.                                                              | 90 playable cards = 86.5% penetration. At ~22 cards per round (20 dealt + ~2 Shine draws), the shoe supports about four rounds before the cut card appears. |
| Shuffle sequence      | Seven riffle shuffles with alternating cuts, a strip, and a final riffle, timed at roughly two minutes for a two-deck shoe. | Meets the minimum mixing requirement; the sequence is logged with the round audit for regulator review.                                                     |
| Dead card during deal | On the third pass, the dealer exposes the 5 of hearts while dealing to BOB.                                                 | The 5 of hearts is placed in the discard rack as a dead card and BOB receives the next card from the shoe. The round continues; no redeal.                  |
| Burn card             | First round of a new shoe: one card is burned face-down to the discard rack before antes are collected.                     | The burned card plays no part in any round of that shoe.                                                                                                    |

## 8. How to Play - Phase-by-Phase Player Guide

Each round follows seven phases. The player guide below describes what happens, what actions are legal, and what the dealer does in each phase, followed by the strategy note used by the in-game coach.

### Phase 1 - Private Reveal

| Aspect        | Detail                                                                                                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------|
| What happens  | Four cards are dealt face-down to each seat. Only you can see your own cards.                                            |
| Your options  | No action required. Review your hand and form a plan.                                                                    |
| Dealer's role | Dealer has no role beyond the deal.                                                                                      |
| Coach tip     | Assess your best two-card pairing and classify it: Strong (350+), Solid (250-349), Marginal (180-249), Weak (below 180). |

### Phase 2 - Showing

| Aspect        | Detail                                                                                                                                                                          |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What happens  | You may turn 0-4 of your cards face-up on the table. Shown cards are visible to every seat for the rest of the round.                                                           |
| Your options  | Toggle any combination of face-down cards face-up, then confirm. Showing ANY card permanently locks out Bow Out.                                                                |
| Dealer's role | Dealer announces and records shown cards.                                                                                                                                       |
| Coach tip     | Strong hands often show one card to deter Spits; weak hands hide everything to keep Bow Out available. Showing is a commitment - never show a hand you are not willing to play. |

### Phase 3 - Bow Out

| Aspect        | Detail                                                                                                                                                             |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What happens  | If you showed no cards, you may forfeit the round and recover 25% of your ante (25 chips on a 100-chip ante).                                                      |
| Your options  | Bow Out or stay. Bowing seats are removed from the round; their remaining pot contribution stays.                                                                  |
| Dealer's role | Dealer processes the refund from the pot and mucks the cards.                                                                                                      |
| Coach tip     | Break-even math: staying costs 100, Bowing costs 75. Stay only if your win probability exceeds about 6.7%. Hands below roughly 150 points should normally Bow Out. |

### Phase 4 - Spit

| Aspect       | Detail                                                                                                                           |
|--------------|----------------------------------------------------------------------------------------------------------------------------------|
| What happens | Each active non-dealer seat receives one Spit: give one face-down card to a target and take one of the target's face-down cards. |

| Aspect        | Detail                                                                                                                                                                                                                                  |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Your options  | Choose your card to give and your target. Each target can be spat at once per round. You may also pass.                                                                                                                                 |
| Dealer's role | Dealer executes the swap, reveals the received card to the spitter only, and marks the target as spat.                                                                                                                                  |
| Coach tip     | Spit dumps your weakest card and draws a random replacement (expected value about 81 points). Simulations show Spitting is worth roughly 9 chips per round versus never Spitting. Strong hands (350+) usually pass to avoid disruption. |

## Phase 5 - Shine

| Aspect        | Detail                                                                                                                                   |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------|
| What happens  | You may discard 0-1 face-down cards and draw replacements from the shoe. The dealer may discard 0-2.                                     |
| Your options  | Choose a face-down card to discard, or keep your hand. You draw only if you discard.                                                     |
| Dealer's role | Dealer discards and draws for each seat, then for the dealer seat.                                                                       |
| Coach tip     | Below 180 points, Shine is usually correct. Above 250, keep. The dealer's extra discard is the house's structural edge - plan around it. |

## Phase 6 - Selection

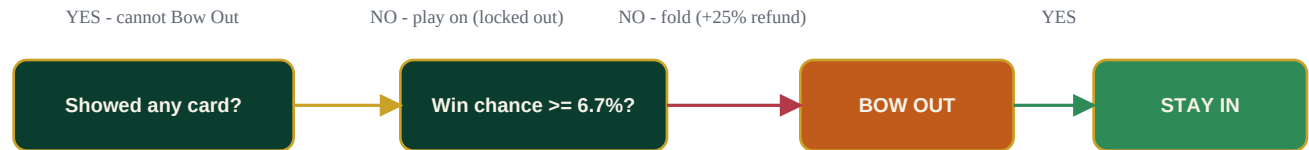
| Aspect        | Detail                                                                                                               |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| What happens  | From your four cards, select the two-card pairing you will play at showdown.                                         |
| Your options  | Choose any two cards. The selection is final.                                                                        |
| Dealer's role | Dealer selects the dealer pair.                                                                                      |
| Coach tip     | Always consider all six pairings. The engine recommends the maximum score; remember Aces count 100 or 0 to maximize. |

## Phase 7 - Showdown & Payout

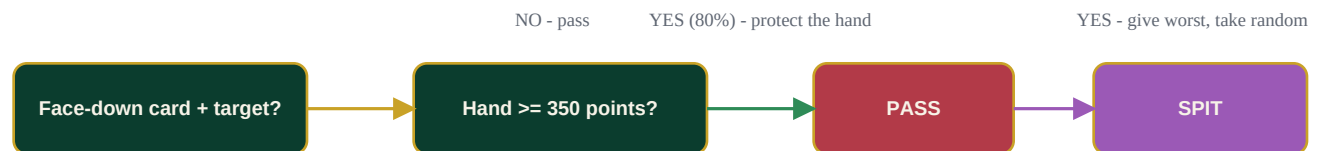
| Aspect        | Detail                                                                                                         |
|---------------|----------------------------------------------------------------------------------------------------------------|
| What happens  | All remaining seats reveal their pairs. The highest score wins; ties split.                                    |
| Your options  | No player action - results are announced.                                                                      |
| Dealer's role | Dealer reveals first, then each seat; announces scores, applies tie-breaks, rakes 25%, and pays the winner(s). |
| Coach tip     | Watch the dealer's revealed pair every round - it is the single most informative datum at the table.           |

## Decision guides (reference cards)

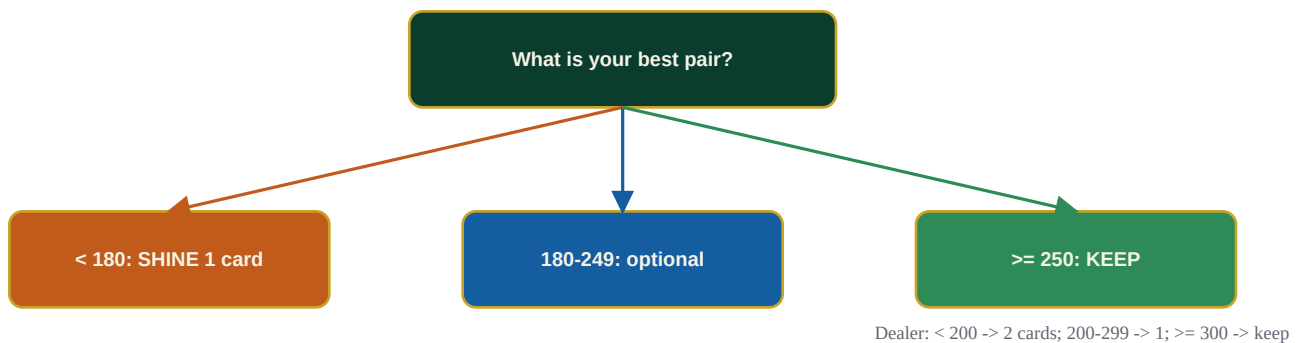
### Bow Out decision



## Spit decision



## Shine decision



## 9. Roles & Responsibilities

A Spit Shine table runs on three human roles: the Player, the Dealer, and the Pit Manager (shift supervisor). The rules are identical for every seat; the roles define who does what, who is responsible for what, and what each role must watch for to keep the game fair. The Pit Manager's watchlist is the game's primary defense against cheating, collusion, and procedural error.

### Responsibility matrix

| Duty                            | Player        | Dealer     | Pit Manager   |
|---------------------------------|---------------|------------|---------------|
| Post ante within limits         | Primary       | Verify     | Set limits    |
| Shuffle / shoe / cut card       | -             | Primary    | Observe       |
| Deal and phase execution        | -             | Primary    | Spot-check    |
| Announce shown cards and scores | -             | Primary    | Audit log     |
| Rake and payouts                | -             | Primary    | Verify + drop |
| Dispute resolution              | Rights        | Escalate   | Final         |
| Fairness monitoring             | Play by rules | Self-check | Primary       |

### 9.1 The Player

The player's job is to make seven decisions per round under information asymmetry: what to show, whether to stay or fold, whether to exchange a card, whether to redraw, and which two cards to play. The full decision guide is in Section 8.

| Player duty      | Detail                                                                                                                |
|------------------|-----------------------------------------------------------------------------------------------------------------------|
| Session setup    | Buy in at least 20x the table ante; choose an ante within limits; confirm the table minimums and the mode before play |
| Hand protection  | Keep all cards above the table and in view; never touch another seat's cards or the pot                               |
| Phase discipline | Decide Showing, Bow Out, Spit, Shine, and Selection only when invited; a confirmed action is final                    |
| Information      | The Spit-received card is private to you; sharing it with other players is against house rules                        |
| Complaints       | Never argue at the table; call the Pit Manager or request the audit log                                               |

Player example: a \$1,000 buy-in at a 100-chip ante covers the recommended 20-round cushion. Two wins, six losses, and two Bow Outs over ten rounds nets -236 chips (worked in Section 12.6).

### 9.2 The Dealer

The dealer executes the game under Pit Manager observation. Full dealing procedures are in Section 7 and payout math in Section 10; this subsection states the dealer's standing duties and self-checks.

| Dealer duty       | Detail                                                                                                                                     |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Equipment control | Verify 52 cards per deck, inspect for marks, shuffle seven riffles + strip + final riffle, insert the cut card, burn one card per new shoe |
| Ante control      | Verify every ante against the published limits before dealing; announce the pot total                                                      |

| Dealer duty      | Detail                                                                                                                                                                   |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deal control     | Round-robin one card at a time; never expose a card; handle dead cards per Section 16.2                                                                                  |
| Phase control    | Announce each phase; confirm shows; process Bow Out refunds (25% of ante, floored); execute Spit swaps without exposing the given card; control Shine discards and draws |
| Payout control   | Compute rake (25% of pot), apply tie-breaks, announce scores, pay winners, and move the rake to the drop box before any other action                                     |
| Audit discipline | Never settle a payout or dispute without the audit log; count the tray at shift start and end                                                                            |

Dealer self-checks every round: card count per seat, face-down integrity during Spit, one target per Spit, discard limits (players 0-1, dealer 0-2), and pot = antes minus refunds. Any discrepancy stops the round and summons the Pit Manager.

## 9.3 The Pit Manager

The Pit Manager owns fairness, money flow, and dispute resolution. The manager inspects the table at open and close, observes shuffles and payouts, reviews the audit log, controls the drop box, and maintains the published limits.

### What can make the game unfair - the watchlist

| Category         | Watch for                                                                                                                                    | Why it matters                                 |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Equipment        | Marked, bent, or trimmed cards; missing cards; duplicate mismatches; patterned or insufficient shuffles; misplaced cut card                  | Compromised randomness or known-card play      |
| Player collusion | Two seats repeatedly Spitting at each other; players sharing private Spit information; signaling; chip passing or soft play between partners | Transfers EV and leaks private information     |
| Betting abuse    | Past posting (adding chips after the result); late antes; ante jumps correlated with deck composition; oversized or phantom bets             | Direct theft and advantage betting             |
| Card handling    | Face-down cards touched or re-arranged; cards mucked early; hands altered after reveal                                                       | Card switching and hand manipulation           |
| Dealer errors    | Payout or rake mistakes; exposed cards; wrong discard counts; Spit swaps that reveal the given card; unannounced shown cards                 | Unfair or uncertified play, disputes           |
| Procedural       | Mid-round seat changes; antes outside limits; missed burn or cut-card reshuffle; unlogged payouts                                            | Breaks the certified procedure and audit trail |

### Phase-by-phase pit watch

| Moment      | What the Pit Manager watches                                                                                                                                   |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before deal | Shuffle and cut; ante limits; roster matches the seated players; shoe position                                                                                 |
| Deal        | Card counts; exposures; deal order; burn card                                                                                                                  |
| Showing     | Announced cards match the physical cards; no re-hiding after confirmation                                                                                      |
| Bow Out     | Refunds equal 25% of each ante; pot updated; mucked cards untouched                                                                                            |
| Spit        | Swap executes without exposing the given card; the received card is shown only to the spitter; one target per round; repeated same-pair Spitting across rounds |
| Shine       | Discard limits (0-1, dealer 0-2); a draw only after a discard; no shoe fishing                                                                                 |
| Selection   | No changes after confirmation                                                                                                                                  |

| Moment            | What the Pit Manager watches                                                       |
|-------------------|------------------------------------------------------------------------------------|
| Showdown & payout | Scores and tie-breaks announced; rake to drop; payout correct; audit entry written |

Pattern cues for surveillance: the same two seats exchanging Spits round after round, a player who always Bows Out when a partner stays, chip movements that do not match payouts, and antes that jump exactly when the shoe runs long.

9.4 Suspected-problem procedure

- Observe and document; never accuse a player at the table.
- Let the round finish, then notify surveillance and the floor.
- Review the audit log and the cameras; decide by evidence.
- Apply house policy (warning, exclusion, or regulator report) without disturbing the game.



## 10. Dealer Procedures & Payout Math

### 9.1 Pot accounting

The pot is the sum of all antes minus Bow Out refunds. At showdown the house rakes 25% (floor of the integer division) and the remainder is paid to the winner, or split among tied winners. Any chips left over after an equal split (for example, splitting 375 two ways leaves 1 chip) are retained by the house as rounding rake - this rule must be posted at the table.

### 9.2 Payout examples

| Scenario                  | Pot | House rake (25%) | Winner(s)    | Payout                       |
|---------------------------|-----|------------------|--------------|------------------------------|
| Full table, no Bow Outs   | 500 | 125              | One winner   | 375                          |
| One Bow Out (25 refund)   | 475 | 118              | One winner   | 357                          |
| Two Bow Outs (50 refunds) | 450 | 112              | One winner   | 338                          |
| Exact tie, full table     | 500 | 125              | Two winners  | 187 each + 1 rounding rake   |
| Cooperative, players win  | 400 | 100              | Four players | 75 each (300 total)          |
| Cooperative, dealer wins  | 400 | -                | Dealer       | House keeps entire pot (400) |

Rounding rake note:  $500 \times 25\% = 125$ , leaving 375. Splitting 375 between two winners yields  $187 + 187 + 1$  remainder, which is retained by the house and logged. All payouts must be announced before chips move.

### 9.3 Dealer's own play

The dealer is a full seat in competitive mode (posts an ante like any seat - 100 chips in the audit model - and can win the pot) but never Bows Out and never Spits. The dealer's structural advantage is the two-card Shine: in 20,000 simulated rounds the extra discard is worth roughly 6.8 points of win rate, lifting the dealer to a 24.6% win rate versus a fair 20%. Dealers follow the same thresholds as the reference AI: Shine 2 cards below 200 points, 1 card at 200-299, keep at 300+.

### 9.4 Disputes and corrections

- Any exposed card during the deal is a dead card: discard it and draw a replacement for that seat.
- If a player exposes a face-down card during Spit or Shine, the card is still played (it becomes known information); the floor decides penalties per house policy.
- Payout disputes are resolved from the audit log; every round's deal, show, bow, spit, shine, selection, and payout events are recorded with card-level detail.

### 9.5 Worked payouts with chip values

| Chip value | Ante  | Pot after one Bow Out | Rake (25%)           | Winner payout        |
|------------|-------|-----------------------|----------------------|----------------------|
| \$0.25     | \$25  | 475 chips = \$118.75  | 118 chips = \$29.50  | 357 chips = \$89.25  |
| \$1.00     | \$100 | 475 chips = \$475.00  | 118 chips = \$118.00 | 357 chips = \$357.00 |
| \$5.00     | \$500 | 475 chips = \$2,375   | 118 chips = \$590    | 357 chips = \$1,785  |

Worked split example at \$1 chips: full table, two winners split 375. Each winner receives 187 chips (\$187); the remaining 1 chip (\$1) goes to the house as rounding rake and is recorded in the audit log.

# 11. Limits, Bankroll & Money Management

Tables do not have fixed stakes: the ante is variable per seat within the 1-500 chip range, and players may drop in or out between rounds. This section defines the chip denominations, the table limits, the buy-in and bankroll guidance, and the cash and chip controls a floor operation needs.

## 11.1 Chip denominations and ante limits

| Denomination | Value at \$1/chip | Typical use         | Buy-in guidance (20x ante)          |
|--------------|-------------------|---------------------|-------------------------------------|
| 1 chip       | \$1               | Minimum ante unit   | 20 chips = \$20 at ante 1           |
| 5 chips      | \$5               | Low-stakes play     | 100 chips = \$100 at ante 5         |
| 25 chips     | \$25              | Standard low limit  | 500 chips = \$500 at ante 25        |
| 100 chips    | \$100             | Standard table ante | 2,000 chips = \$2,000 at ante 100   |
| 500 chips    | \$500             | High limit          | 10,000 chips = \$10,000 at ante 500 |

Ante limits: minimum 1 chip, maximum 500 chips per seat. The house publishes its table minimum and maximum inside this range; each seat chooses its ante freely within those limits each round. The maximum single-round win is the full pot minus rake - at a full table at 100-chip antes that is 375 chips. A player can never win more than the pot, so no separate jackpot exposure exists unless a future side wager is added (which would require its own certification).

### Variable table rules

- Players may join or leave between rounds; seats never change mid-round, and empty seats post no ante and take no cards.
- Table capacity is four player seats plus the dealer; the table publishes its minimum player count (one for heads-up).
- Ante spread limit (recommended): the largest and smallest ante at one table should stay within a 5x spread. Mixed antes transfer EV toward smaller bettors (Section 14.9), so an unbounded spread is bad for high-ante players and for the game's perceived fairness.

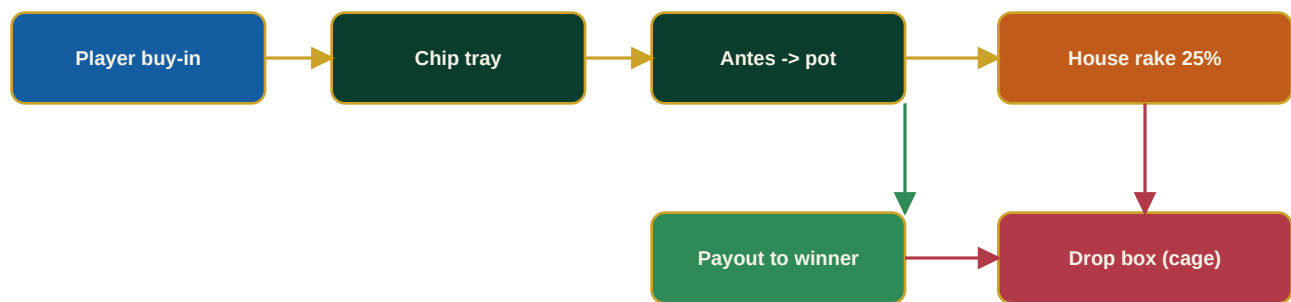
## 11.2 Player bankroll guidance

| Guideline           | Rule of thumb                                                                            | Worked example (\$1 chips)                                                                      |
|---------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Session buy-in      | At least 20x the ante                                                                    | 100-chip ante -> \$2,000 buy-in; 25-chip ante -> \$500                                          |
| Session loss limit  | Stop at 5x the ante                                                                      | Stop-loss at 5x your ante per session                                                           |
| Bow Out discipline  | Fold below ~150 points when hidden                                                       | Saves 25 chips per fold; at ~9% of rounds this adds ~2.3 chips per round versus never folding   |
| Expected round cost | Audited EV scales with ante: about -0.34 chips per chip wagered at a full table (seat 0) | At a 100-chip ante, EV -33.9/round; a \$2,000 bankroll supports roughly 50-60 rounds on average |

## 11.3 Cash and chip flow

Funds move in a closed loop: player buy-ins exchange cash for chips from the tray; antes move chips to the pot; at showdown the rake moves to the drop box and the payout moves back to the winner's stack; Bow Out refunds return to the player. At end of shift the chip equation must balance.

| Reconciliation Item     | Amount                                  | Worked example                                                                                                |
|-------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Opening float (tray)    | +\$2,000                                | Chips issued to the table                                                                                     |
| Player buy-ins (cash)   | +\$4,000                                | Cash exchanged for chips                                                                                      |
| Closing tray            | -\$2,100                                | Chips still in the tray                                                                                       |
| Drop box (cash removed) | -\$3,900                                | Cash moved to the cage                                                                                        |
| Check                   | $\$2,000 + \$4,000 = \$2,100 + \$3,900$ | Balanced: $\$6,000 = \$6,000$                                                                                 |
| Audit theoretical win   | \$1,100                                 | Sum of per-round rakes recorded by the audit log                                                              |
| Variance monitoring     | +2.9%                                   | Actual win \$1,132 vs theoretical \$1,100 - within the standard +/-3% tolerance; investigated only if outside |



Bow Out refunds return 25% of ante to the player before rake is computed

## 11.4 Responsible gaming and jurisdictional limits

- Publish table minimums and maximums at the table and in the house rules; they are configurable per jurisdiction and per venue.
- Support player-set loss limits, time limits, and voluntary exclusion through the standard responsible-gaming tools.
- Currency transaction reporting and AML thresholds apply at the cage, not the table; the dealer never handles cash directly - all buy-ins are exchanged through the tray under surveillance.

## 12. Daily Turnover & Revenue Model

This section models the funds that flow through a Spit Shine table and the revenue it can generate. All figures are illustrative and depend on four operator-set variables: chip value, round duration, table occupancy, and operating hours. The audit supplies the fixed part: the house keeps 24.5% of the pot (rake) in competitive mode.

### 12.1 Definitions

| Term                    | Definition                                      | Spit Shine value                                                                  |
|-------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------|
| Handle (gross turnover) | Total antes placed across all seats in a period | Audited average 565 chips per round under the variable-table model (Section 14.9) |
| Table win (hold)        | House revenue before expenses                   | Rake only: 24.5% of pot in competitive mode                                       |
| Hold percentage         | Table win / handle                              | 24.5% audited, competitive, at any table size or ante                             |
| Drop                    | Cash and chips exchanged at the table           | Tracked per shift; reconciled against theoretical win                             |
| Rounds per hour         | Dealt rounds per hour of operation              | Model range 17-30 (see assumptions)                                               |

### 12.2 Model assumptions (must be validated in floor trials)

| Variable              | Conservative               | Standard                   | Aggressive                 |
|-----------------------|----------------------------|----------------------------|----------------------------|
| Round duration        | 3.5 minutes                | 2.5 minutes                | 2.0 minutes                |
| Rounds per hour       | 17                         | 24                         | 30                         |
| Operating hours / day | 12                         | 12                         | 24                         |
| Table model           | Variable-table audit model | Variable-table audit model | Variable-table audit model |
| Chip value            | \$1                        | \$1                        | \$1                        |
| House win per round   | \$138.40 (audited average) | \$138.40                   | \$138.40                   |

### 12.3 Per-round economics

| Configuration                                                  | Handle per round  | Expected house win         | Winner payout               |
|----------------------------------------------------------------|-------------------|----------------------------|-----------------------------|
| Variable table (audited model, avg 3.59 players, mixed antes)  | 565 chips = \$565 | \$138.40 (24.5%)           | Varies with pot             |
| Competitive, full table at 100-chip antes (4 players + dealer) | 500 chips = \$500 | \$122.50 (24.5%)           | 375 chips = \$375           |
| Solo heads-up at 100-chip antes (1 player + dealer)            | 200 chips = \$200 | \$49.40 (24.7%)            | 150 chips = \$150           |
| Competitive, 3 players + dealer                                | 400 chips = \$400 | \$98.00 (24.5%)            | 300 chips = \$300           |
| Cooperative, full table (4 players, dealer antes 0)            | 400 chips = \$400 | \$184.40 (46.1% - flagged) | 75 chips per winning player |

## 12.4 Daily turnover scenarios (worked)

| Scenario                    | Rounds per day | Handle                  | House win (24.5%)            |
|-----------------------------|----------------|-------------------------|------------------------------|
| Conservative: 17 rph x 12 h | 204            | 204 x \$565 = \$115,260 | \$115,260 x 24.5% = \$28,215 |
| Standard: 24 rph x 12 h     | 288            | 288 x \$565 = \$162,720 | \$162,720 x 24.5% = \$39,833 |
| Aggressive: 30 rph x 24 h   | 720            | 720 x \$565 = \$406,800 | \$406,800 x 24.5% = \$99,582 |

These figures use the audited variable-table model directly: the average table fields 3.59 player seats with mixed 1-500 chip antes, so no separate occupancy factor is needed. Handle scales linearly with average ante: at a 25-chip average ante, divide by 4; at a 500-chip average ante, multiply by 5.

## 12.5 Monthly and multi-table projections

| Deployment          | Handle / day | House win / day | House win / 30 days |
|---------------------|--------------|-----------------|---------------------|
| 1 table, standard   | \$162,720    | \$39,833        | \$1,194,990         |
| 5 tables, standard  | \$813,600    | \$199,165       | \$5,974,950         |
| 10 tables, standard | \$1,627,200  | \$398,330       | \$11,949,900        |

These projections assume the model's standard case (24 rounds/hour, 12 hours/day) and full-year operation at the modelled drop-in/out roster. They are deliberately striking because a 25% rake is far above mainstream table games (blackjack hold is typically 10-25% of drop but with much lower edge per bet; baccarat holds 10-15%). The rake is the game's key commercial lever: Section 12.7 shows calibration options.

## 12.6 Player session example (funds flow)

| Round | Result           | Player cash flow | Running total |
|-------|------------------|------------------|---------------|
| 1     | Win (full table) | +257             | +257          |
| 2     | Loss             | -100             | +157          |
| 3     | Bow Out          | -75              | +82           |
| 4     | Loss             | -100             | -18           |
| 5     | Win (full table) | +257             | +239          |
| 6     | Loss             | -100             | +139          |
| 7     | Bow Out          | -75              | +64           |
| 8     | Loss             | -100             | -36           |
| 9     | Loss             | -100             | -136          |
| 10    | Loss             | -100             | -236          |

Session summary: \$1,000 buy-in, 10 rounds, 2 wins, 6 losses, 2 Bow Outs. Net = 2 x (+257) + 6 x (-100) + 2 x (-75) = +514 - 600 - 150 = -236 chips (-\$236 at \$1 chips). This is an illustrative sequence, not a prediction.

## 12.7 Calibration options for market fit

| Rake          | Hold (approx.) | House win per full round | Note                                         |
|---------------|----------------|--------------------------|----------------------------------------------|
| 25% (current) | 24.5%          | \$122.50                 | Audited baseline; very high for a table game |

| Rake | Hold (approx.) | House win per full round | Note                                    |
|------|----------------|--------------------------|-----------------------------------------|
| 15%  | 14.7%          | \$73.50                  | Closer to premium table-game norms      |
| 10%  | 9.8%           | \$49.00                  | Competitive with popular carnival games |

The operator chooses the rake and chip value as a package. A lower rake improves player retention and table longevity; a higher chip value restores revenue per table. Both variables must be settled before the independent game-math certification, because they change the certified payout tables.

## 12.8 Round-time and pace model

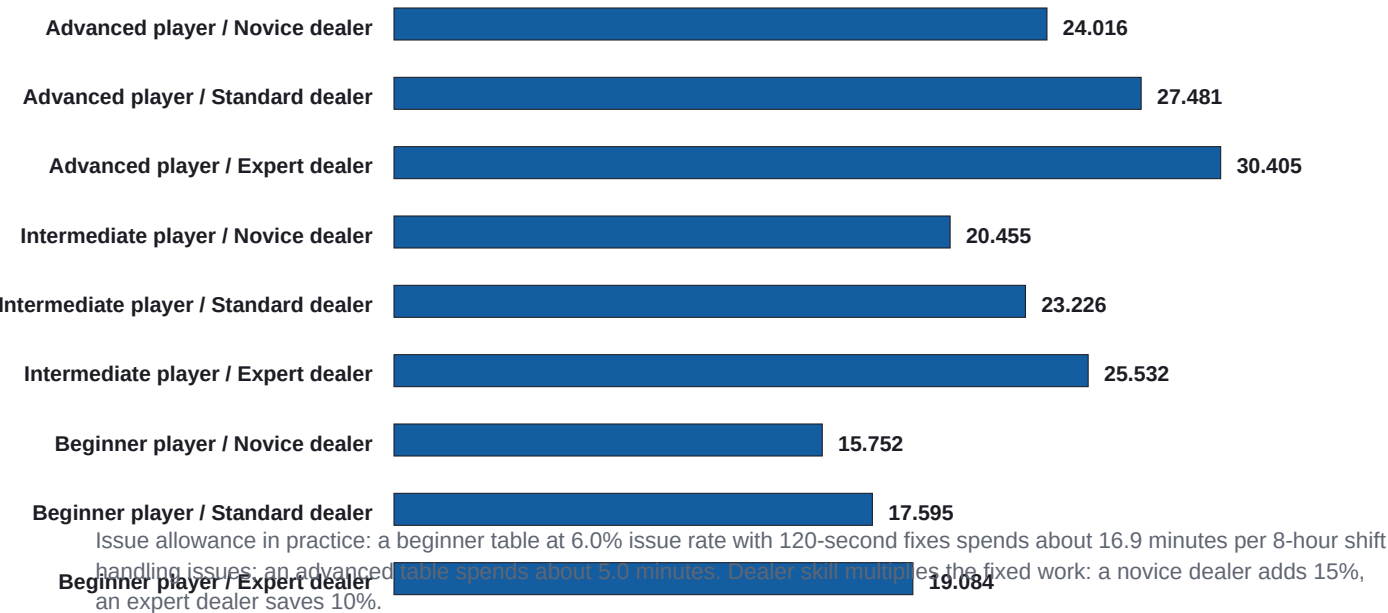
Round time is the product of three components: the dealer's fixed procedural work (shuffle exposure, deal, announcements, Spit and Shine administration, showdown and payout), the players' decision time (scaled by skill), and an allowance for explaining rules and fixing mid-game issues (misdeals, disputes, questions). The model below is the basis for the 24-hour distribution in Section 14.12.

| Component                                   | Advanced                            | Intermediate                         | Beginner                              |
|---------------------------------------------|-------------------------------------|--------------------------------------|---------------------------------------|
| Player decision time per round              | 56.000 s                            | 70.000 s (x1.250)                    | 89.600 s (x1.600)                     |
| Explanation + issue-fix allowance           | 5.000 s (1.5% issue rate, 90 s fix) | 15.000 s (3.0% issue rate, 90 s fix) | 45.000 s (6.0% issue rate, 120 s fix) |
| Dealer fixed + admin time (standard dealer) | 70.000 s                            | 70.000 s                             | 70.000 s                              |

### Round time and rounds per hour by configuration

| Configuration                         | Average round time | Rounds per hour |
|---------------------------------------|--------------------|-----------------|
| Advanced player / Novice dealer       | 149.900 s          | 24.016 rph      |
| Advanced player / Standard dealer     | 131.000 s          | 27.481 rph      |
| Advanced player / Expert dealer       | 118.400 s          | 30.405 rph      |
| Intermediate player / Novice dealer   | 176.000 s          | 20.455 rph      |
| Intermediate player / Standard dealer | 155.000 s          | 23.226 rph      |
| Intermediate player / Expert dealer   | 141.000 s          | 25.532 rph      |
| Beginner player / Novice dealer       | 228.540 s          | 15.752 rph      |
| Beginner player / Standard dealer     | 204.600 s          | 17.595 rph      |
| Beginner player / Expert dealer       | 188.640 s          | 19.084 rph      |

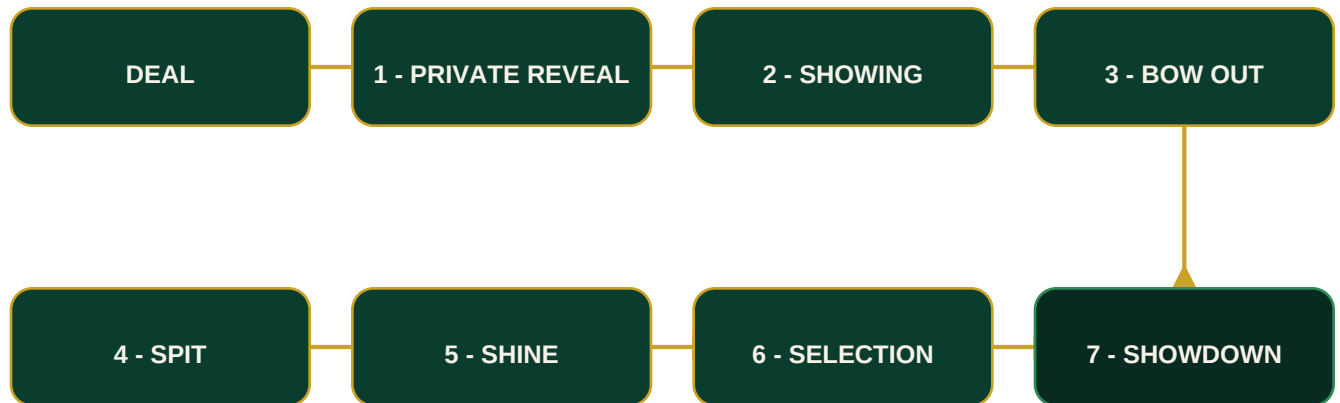
Rounds per hour by player/dealer skill (includes explanation and issue-fix time)



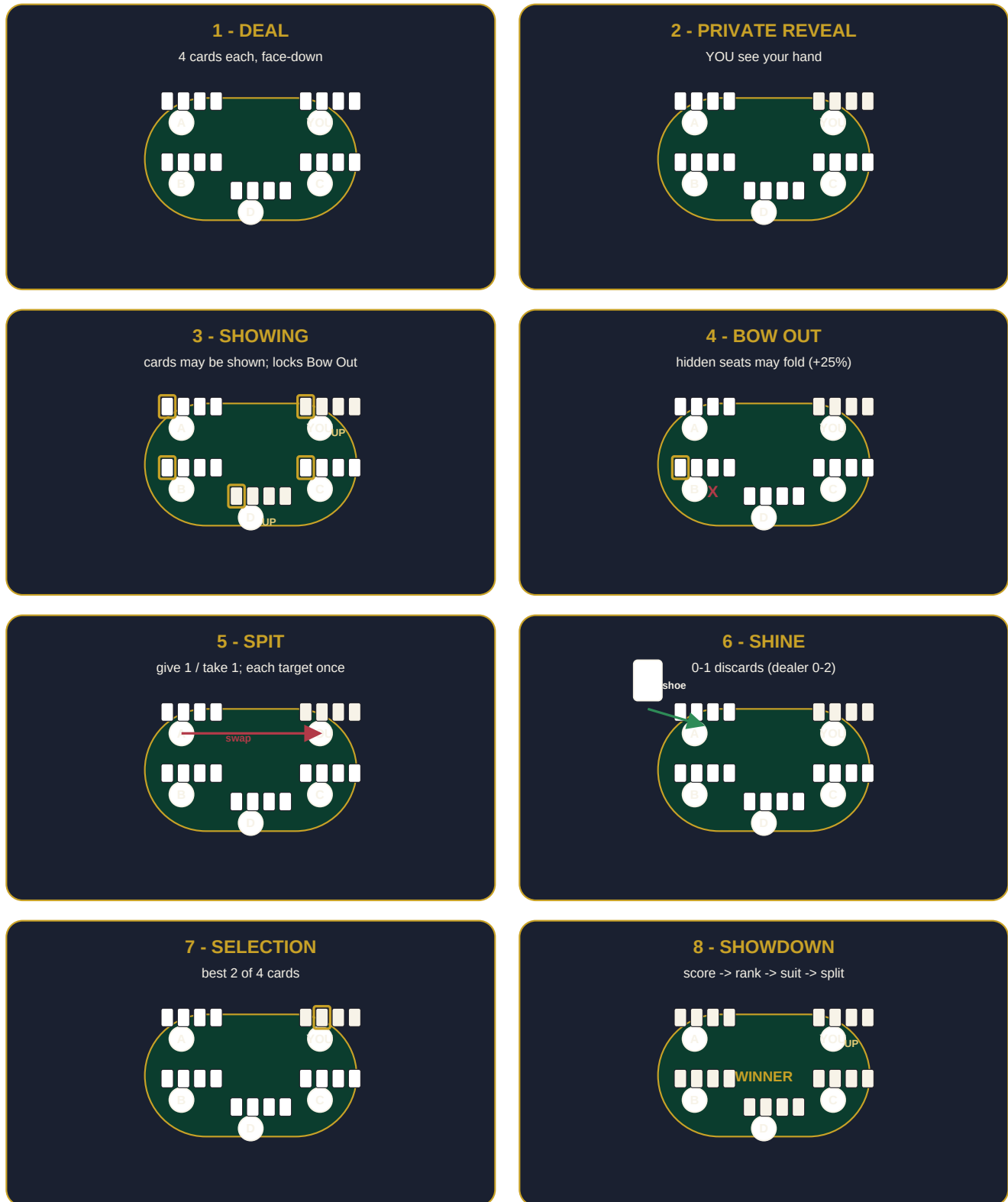


## 13. Game State Machine & Canonical Rules

### Round state flow



### The table in every state



Each panel shows the table state at one moment of the round: face-down cards are dark, face-up cards are light, gold rings mark shown or selected cards, X marks a Bow Out, and arrows show card movement. The same states are reflected in the audit log after every round.

## Value flow through the round



Example: 100-chip ante, 5 seats, one Bow Out (25 refund). All values in chips, 3 decimal places.

The pot is the only value that moves through the round: antes create it, Bow Out refunds reduce it, and the rake splits it at showdown. Spit and Shine move cards, never chips, so the pot is constant through Phases 4-6 unless a Bow Out refund was processed in Phase 3. Rounding rake (e.g., 118.750 -> 118) stays with the house and is logged.

## Canonical rules reference (v11)

| Rule      | Specification                                                                         |
|-----------|---------------------------------------------------------------------------------------|
| Shoe      | 104 cards (2 decks), fresh and shuffled every round                                   |
| Deal      | 4 cards per seat, round-robin, one card at a time                                     |
| Ante      | 100 per seat; dealer antes 0 in cooperative mode                                      |
| Showing   | 0-4 cards; showing any card locks out Bow Out                                         |
| Bow Out   | Hidden seats only; refund 25% of ante; dealer never bows                              |
| Spit      | One per non-dealer seat; each target once per round; received card private to spitter |
| Shine     | 0-1 discards (players), 0-2 (dealer); draw only after discard                         |
| Selection | Best 2 of 4; human may choose any two cards                                           |
| Showdown  | Score, then high rank, then high suit; exact ties split                               |
| Payout    | Rake 25% of pot; winners share remainder; rounding rake to house                      |
| Modes     | Competitive (all seats) and Cooperative (players vs dealer)                           |

## State invariants

- A seat can Bow Out only once and only while hidden.
- A seat can Spit at most once; a target can receive at most one Spit.
- Spit and Shine operate only on face-down cards.
- The pot never goes negative; refunds are taken from the pot before rake.
- The dealer is always active at showdown (never Bows Out).
- Every action is written to the audit log before the next action begins.

## 14. Audit Data - Odds, EV & Fairness

All figures below are produced by the canonical engine v11 and the audit harness (spitshine\_analysis.py). Exact probabilities are enumerated combinatorially; round-level figures come from 20,000 Monte Carlo rounds per scenario with a fixed seed (20260804), playing the documented threshold policies at every seat. Chip conservation (sum of seat nets + house take = 0) holds to machine precision.

### 14.1 Exact probabilities

| Event                                                | Probability | Method                                 |
|------------------------------------------------------|-------------|----------------------------------------|
| A specific 2-card pair is A+A                        | 0.523%      | Exact - 28 of $C(104,2) = 5,356$ pairs |
| Your 4-card hand contains A+A                        | 2.895%      | Exact hypergeometric                   |
| At least one seat dealt A+A in a round               | 13.917%     | Monte Carlo - 200,000 deals            |
| At least one seat finishes with A+A after Spit/Shine | 17.210%     | 20,000-round simulation                |
| Best pair 350+ (any seat's deal)                     | 11.501%     | Exact hand enumeration                 |

The Spit and Shine phases create roughly 3.3 points of additional Double-Aces frequency because a lone Ace can be paired by a redraw or exchange. This is a deliberate feature of the game's variance profile.

### 14.2 Round outcomes - competitive mode (20,000 rounds)

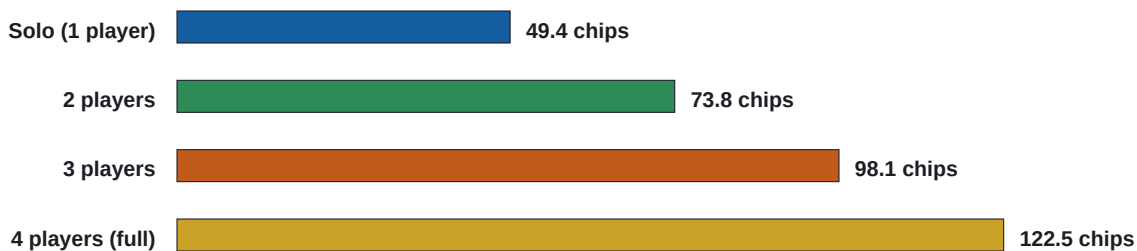
| Seat            | Win rate | Avg final score | Net EV per round |
|-----------------|----------|-----------------|------------------|
| YOU (seat 0)    | 17.830%  | 241.5           | -33.875          |
| ALICE (seat 1)  | 19.330%  | 246.7           | -28.654          |
| BOB (seat 2)    | 20.010%  | 249.0           | -26.529          |
| CAROL (seat 3)  | 21.445%  | 255.3           | -20.981          |
| DEALER (seat 4) | 24.610%  | 278.0           | -12.476          |
| House (rake)    | -        | -               | +122.5 per round |

### Win rate by seat - competitive (fair baseline 20%)



Every seat - including the dealer - has negative expected value in competitive mode; only the house rake is positive. The dealer's 24.6% win rate is the strongest structural advantage in the game, and it comes almost entirely from the two-card Shine.

### House win per round by table size (100-chip ante, audited)



House win scales with the pot: 24.5% of (players + dealer) x ante. The hold percentage is invariant; only the volume changes with table size.

## 14.3 Winner score distribution

| Winning score      | Rounds | Share   |
|--------------------|--------|---------|
| Double Aces (1000) | 3,442  | 17.210% |
| 350-499            | 7,307  | 36.535% |
| 250-349            | 8,777  | 43.885% |
| 180-249            | 474    | 2.370%  |

## 14.4 House edge - corrected mathematics

With a full 500-chip pot and a fair 20% win rate, the winner nets  $375 - 100 \text{ ante} = +275$  chips. Expected value is therefore  $E = 0.20 \times 275 - 0.80 \times 100 = -25$  chips per round. That is a 25% house edge per ante, exactly the 25% rake

- not the 5% claimed in the original white paper, which incorrectly treated the gross 375 payout as profit. Measured house take in simulation is 122.5 chips per round (24.503%) after Bow Out refunds.

| Break-even question         | Correct answer      | Formula                           |
|-----------------------------|---------------------|-----------------------------------|
| Stay in vs Bow Out          | ~6.7% win chance    | 25 / 375                          |
| Break even for the round    | ~26.7% win chance   | 100 / 375                         |
| Fair-game EV (20% win rate) | -25 chips per round | $0.2 \times 275 - 0.8 \times 100$ |

### Worked EV calculations

| Perspective                  | Formula                                 | Result                          | Measured (20k rounds) |
|------------------------------|-----------------------------------------|---------------------------------|-----------------------|
| Player, seat 0               | $0.1783 \times 275 - 0.8217 \times 100$ | -33.1 chips / round             | -33.9 chips / round   |
| Dealer seat                  | $0.2461 \times 275 - 0.7539 \times 100$ | -7.7 chips / round              | -12.5 chips / round   |
| House                        | Rake only (25% of pot)                  | +125.0 chips / round (full pot) | +122.5 chips / round  |
| Bow decision, 12% win chance | $0.12 \times 275 - 0.88 \times 100$     | -55.0 -> stay (bow costs -75)   | -                     |
| Bow decision, 5% win chance  | $0.05 \times 275 - 0.95 \times 100$     | -81.3 -> bow (saves 6.3)        | -                     |

The small gap between formula and measured EV comes from pot-size variation: Bow Out refunds shrink the pot, so the actual winner payout is below the 375-chip full-pot figure used in the closed-form formula.

## 14.5 Cooperative mode economics and rebalancing

In cooperative mode the dealer antes 0, the pot is 400, and the team of four players wins if any player beats the dealer. The team wins 70.3% of rounds in simulation, yet each player still nets -46.140 per round because a team win pays only 75 chips gross per player, and a dealer win forfeits the entire pot. The house takes 184.4 chips per 400-pot (46.096% effective edge). This mode is not viable for a casino floor in its current form and is flagged for rebalancing.

### Rebalance package (illustrative, pending certification)

| Option                                 | Change                                                         | Audited/derived effect                                                                  |
|----------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| A - Reduce dealer Shine in cooperative | Dealer discards 0-1 cards like players                         | Team win rate rises from 70.3% to 78.6% (20,000-round experiment, seed 20260806)        |
| B - Dealer antes + higher team payout  | Dealer antes 100; team win pays 125 gross per player (pot 500) | Player EV improves from -46.1 to about -12.1 chips/round                                |
| C - Combined package                   | Both A and B                                                   | Player EV about -1.8 chips/round - near fair, comparable to standard house-banked games |

Option C is the recommended starting point for a casino deployment: it removes the dealer's double-discard edge in this mode and makes the payout structure fair while keeping the house positive. It must be formally simulated and certified before publication.

### The attraction package (specialty/event use only)

Cooperative mode is repositioned as an optional event attraction, not a standard table. The recommended package pays 140 chips gross per active player on a team win (the house subsidizes 20 chips), reduces the dealer to a one-card Shine, and has the dealer post a full ante.

| Attraction metric                    | Audited value (100,000 rounds) |
|--------------------------------------|--------------------------------|
| Team win rate                        | 78.800%                        |
| Dealer win rate                      | 21.200%                        |
| Player expected value (EV) per round | +3.103 chips                   |
| House cost per round (combined)      | -12.414 chips                  |
| House gross take per round           | +87.586 chips                  |
| Handle per round                     | 500.000 chips                  |

### Conversion economics - why the loss pays for itself

| Item                                                       | Worked value                  |
|------------------------------------------------------------|-------------------------------|
| House cost of one 20-round attraction session (4 players)  | 20 x -12.414 = +248.271 chips |
| House win per competitive full-table round                 | 122.514 chips                 |
| House win per player per competitive round (share of rake) | 30.629 chips                  |
| Competitive player-rounds to recover one session           | 8.106                         |

One 20-round attraction session costs the house about +248.271 chips. Recovering it requires about 8.106 competitive player-rounds, or 2.026 full-table rounds. A single returning player playing a short session usually pays for several events. The attraction is therefore a marketing expense with a measured conversion return, never a standard-table offering.

## 14.6 Rule variants - what each mechanic contributes

| Variant                             | Dealer win | YOU win | YOU net/round |
|-------------------------------------|------------|---------|---------------|
| Baseline (canonical)                | 24.610%    | 17.830% | -33.875       |
| Dealer shines 1 card (like players) | 18.330%    | 18.860% | -30.104       |
| Dealer also Spits (legacy v4 rule)  | 29.720%    | 17.140% | -36.138       |
| Spit order rotates each round       | 24.940%    | 18.895% | -30.264       |
| Human never Spits                   | 26.080%    | 15.330% | -42.879       |
| Nobody Shines                       | 17.800%    | 18.845% | -30.628       |
| Conservative 'smart' human          | 25.090%    | 15.345% | -36.331       |

Reading the variants: the dealer's second Shine discard is worth about 6.8 points of win rate (24.6% to 18.3%); the legacy v4 rule that let the dealer Spit pushed the dealer to 29.7%; rotating the Spit order equalizes the player seats; and never Spitting costs the player roughly 9 chips per round.

## 14.7 Seat-position fairness

With a fixed Spit order, seat 0 (YOU) wins 17.830% while seat 3 (CAROL) wins 21.445% - a systematic positional bias of about 3.7 points. Reversing the order inverts the bias, confirming the Spit sequence itself is the cause. The recommended floor fix is to rotate the Spit starting seat each round (poker-button style), which equalizes the four player seats without changing house economics. This must be included in the table procedures.

14.8 Solo vs dealer (heads-up) odds

The canonical engine seats 1-4 players plus the dealer, making a true heads-up table possible. The player only needs to beat the dealer, so heads-up win rates are far higher than at a full table; the 25% rake still leaves the house positive in competitive mode.

| Metric (100-chip ante, 20k rounds) | Competitive heads-up | Cooperative heads-up | Full table (reference) |
|------------------------------------|----------------------|----------------------|------------------------|
| Player win rate                    | 47.975%              | 43.505%              | 17.830%                |
| Dealer win rate                    | 53.395%              | 56.495%              | 24.610%                |
| Player EV / round                  | -26.815              | -65.157              | -33.875                |
| House win / round                  | +49.4 (24.685% hold) | +65.2 (65.157% hold) | +122.5 per round       |
| Player average final score         | 261.8                | 266.1                | 241.5                  |

Competitive heads-up is the recommended solo configuration. Cooperative heads-up gives the house a 65.2% hold and should be rebalanced or withdrawn (see Section 14.5).

14.9 Variable-table simulation (drop-in/out, mixed antes)

To model a real table, 20,000 rounds were simulated with players dropping in and out between rounds (10% leave probability, 45% join probability, capacity 4) and every seat's ante sampled per round from a mixed 1-500 chip distribution (55% low, 30% mid, 15% high). The dealer posts a fixed 100-chip ante in competitive mode.

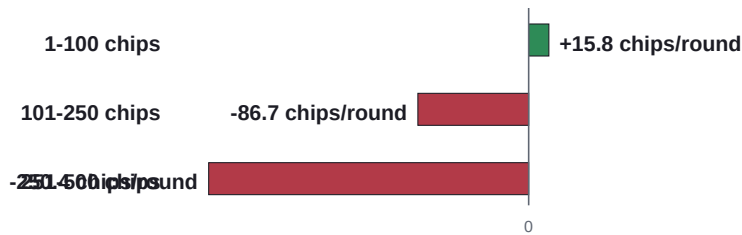
| Metric               | Value             | Note                                                   |
|----------------------|-------------------|--------------------------------------------------------|
| Average player seats | 3.59              | 1p: 1.3% of rounds, 2p: 7.3%, 3p: 22.6%, 4p: 68.8%     |
| Drop-ins / drop-outs | 5,156 / 5,154     | Balanced over 20,000 rounds                            |
| Handle per round     | 565.3 chips       | Mixed antes across 3.59 average seats + dealer         |
| House win per round  | 138.37 chips      | Hold 24.479% - invariant to table size                 |
| Player (YOU) net     | -54.502 per round | Reflects the mixed ante model, not a single ante level |
| Dealer seat net      | +6.074 per round  | Fixed 100 ante; benefits from winning mixed-stake pots |

EV transfer across ante bands (audited)

| YOU ante band | Rounds | Net EV per round |
|---------------|--------|------------------|
| 1-100 chips   | 11,005 | +15.806          |
| 101-250 chips | 6,038  | -86.695          |
| 251-500 chips | 2,957  | -250.425         |



YOU EV by ante band (variable-table audit, 20k rounds)



The winner takes 75% of the whole pot regardless of their own ante, so small bettors at a mixed-stakes table capture outsized pots and can be net positive, while large bettors lose more than their proportional share of the rake. This is a real design consequence of variable stakes: operators should publish an ante spread limit (recommended 5x between the smallest and largest ante at one table) or run uniform-ante tables to protect high-ante players.

14.10 One-million-round verification

To confirm every headline figure at casino-grade sample size, the canonical engine ran 1,000,000 rounds for each of the three flagship scenarios (competitive full table, variable drop-in/out table, and solo heads-up) with audit logging disabled for speed and fixed seeds. Total compute was about 15 minutes. Win rates carry a 95% confidence interval of roughly +/-0.08 to +/-0.10 percentage points.

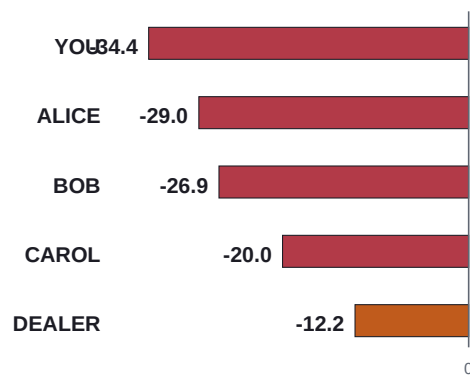
| Scenario       | Metric              | 1M result    | 95% CI     | 20k reference |
|----------------|---------------------|--------------|------------|---------------|
| Competitive    | YOU win rate        | 17.729%      | +/-0.07 pp | 17.830%       |
| Competitive    | Dealer win rate     | 24.657%      | +/-0.08 pp | 24.610%       |
| Competitive    | YOU EV per round    | -34.42 chips | -          | -33.88        |
| Competitive    | House win per round | 122.53 chips | -          | 122.51        |
| Variable table | Average players     | 3.586        | -          | 3.589         |
| Variable table | Hold percentage     | 24.472%      | -          | 24.479%       |
| Variable table | House win per round | 138.27 chips | -          | 138.37        |
| Solo heads-up  | YOU win rate        | 48.485%      | +/-0.10 pp | 47.975%       |
| Solo heads-up  | YOU EV per round    | -25.84 chips | -          | -26.82        |

Every 20,000-round figure survives the million-round check inside the expected sampling error. The verified constants: hold 24.47% (variable table), competitive house win 122.53 chips per round at 100-chip antes, solo player win rate 48.49%, and the seat-order bias of about 4 points between seat 0 and seat 3.

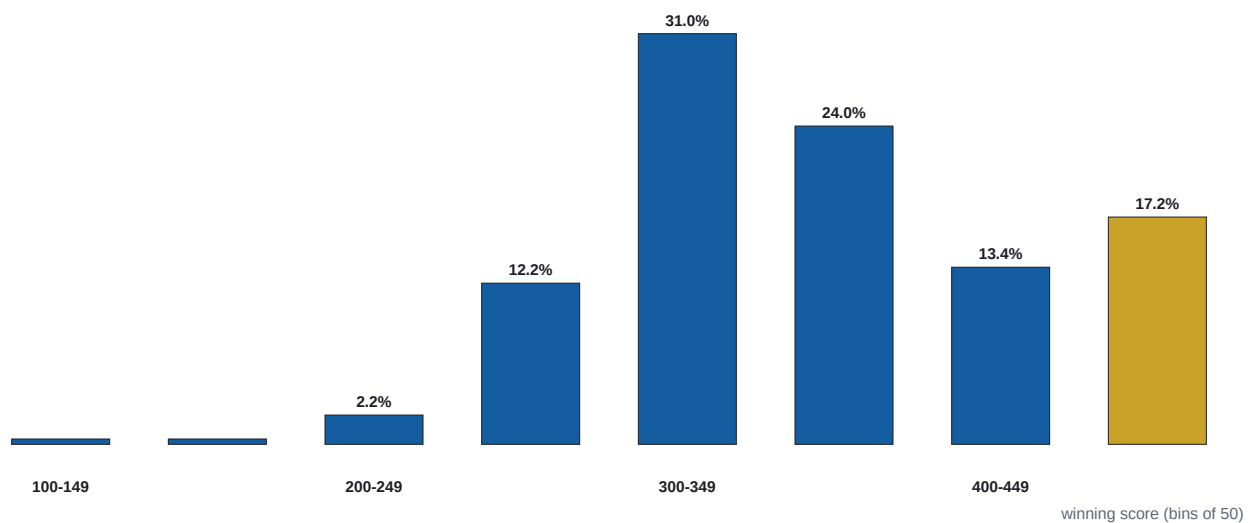
## Win rate by seat - competitive, 1,000,000 rounds



## Net EV per seat per round - competitive, 1,000,000 rounds (100-chip ante)

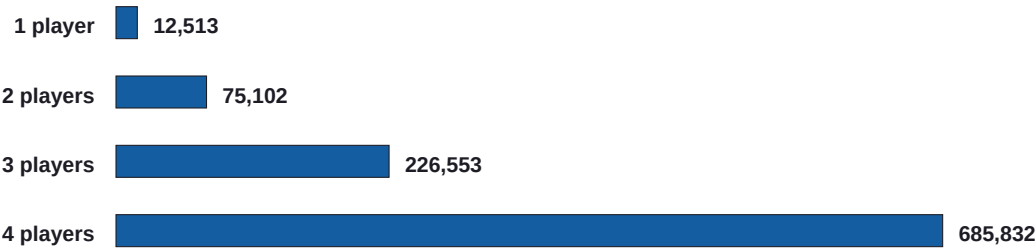


## Winning-score distribution - competitive, 1,000,000 rounds

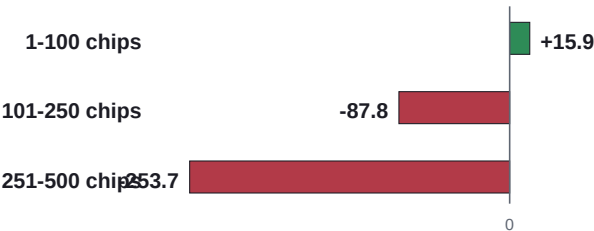


Variable-table metrics - 1,000,000 rounds

Rounds by table size - variable-table model



YOU EV by ante band - variable-table model, 1,000,000 rounds



Solo heads-up win rate - 1,000,000 rounds



The EV transfer across ante bands is confirmed at 1,000,000 rounds: 1-100 chip bettors net +15.9 chips per round while 251-500 chip bettors lose -253.7 - the strongest argument for the 5x ante spread limit.

14.11 Two-million-round hand audit & 24-hour distribution

The canonical engine played 2,000,000 competitive rounds while tracking every seat's hand and point totals per round: the dealt best pair, the final selected pair, and the final score for all 10,000,000 hands. The full-run aggregates are below; a raw 5,000-round sample (every card, every score) is exported to spitshine\_audit\_2m\_sample.csv.

| Audit metric    | Value                                           |
|-----------------|-------------------------------------------------|
| Rounds played   | 2,000,000                                       |
| Hands tracked   | 10,000,000 (8,000,000 player hands)             |
| Total handle    | 1,000,000,000 chips (500 per round at 100-ante) |
| House win total | 245,057,169 chips (122.53 per round)            |

| Audit metric          | Value                                         |
|-----------------------|-----------------------------------------------|
| Hold percentage       | 24.51%                                        |
| Bow Out refunds paid  | 18,179,000 chips                              |
| Table-days equivalent | 3,472.2 standard 576-round days               |
| Raw sample exported   | 5,000 rounds in spitshine_audit_2m_sample.csv |

| Seat   | Avg dealt | Avg final | Delta | AA dealt | AA final | Win rate | Total points (2M rounds) |
|--------|-----------|-----------|-------|----------|----------|----------|--------------------------|
| YOU    | 238.3     | 230.1     | -8.2  | 57,715   | 61,050   | 17.760%  | 482,538,440              |
| ALICE  | 238.2     | 237.2     | -1.0  | 57,634   | 70,150   | 19.263%  | 495,242,140              |
| BOB    | 238.3     | 239.1     | +0.9  | 57,864   | 71,737   | 19.794%  | 498,677,050              |
| CAROL  | 238.0     | 245.8     | +7.8  | 57,197   | 78,321   | 21.657%  | 510,213,430              |
| DEALER | 238.4     | 267.6     | +29.2 | 58,193   | 76,198   | 24.660%  | 554,204,580              |

All seats are dealt identically (avg ~238 points); the dealer's two-card Shine is visible as the +29-point final-score lift. The human seat finishes lower (-8) because the default policy Bows Out weak hands, which protects bankroll but lowers the average hand played.

Hand-score tiers - dealt vs final, 2,000,000 rounds (all seats)

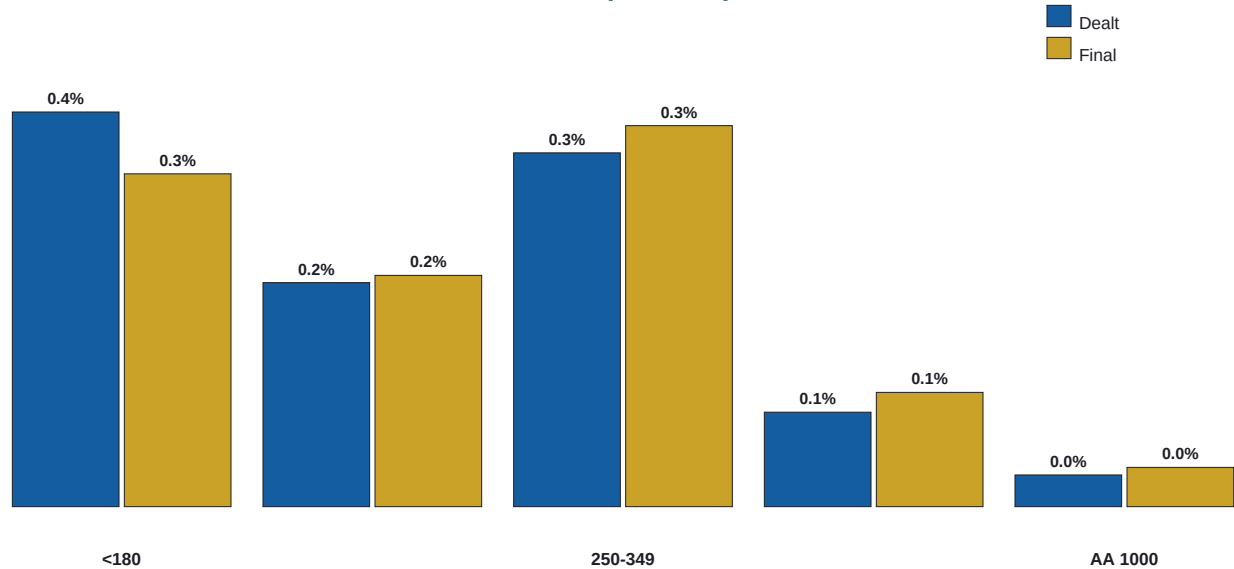
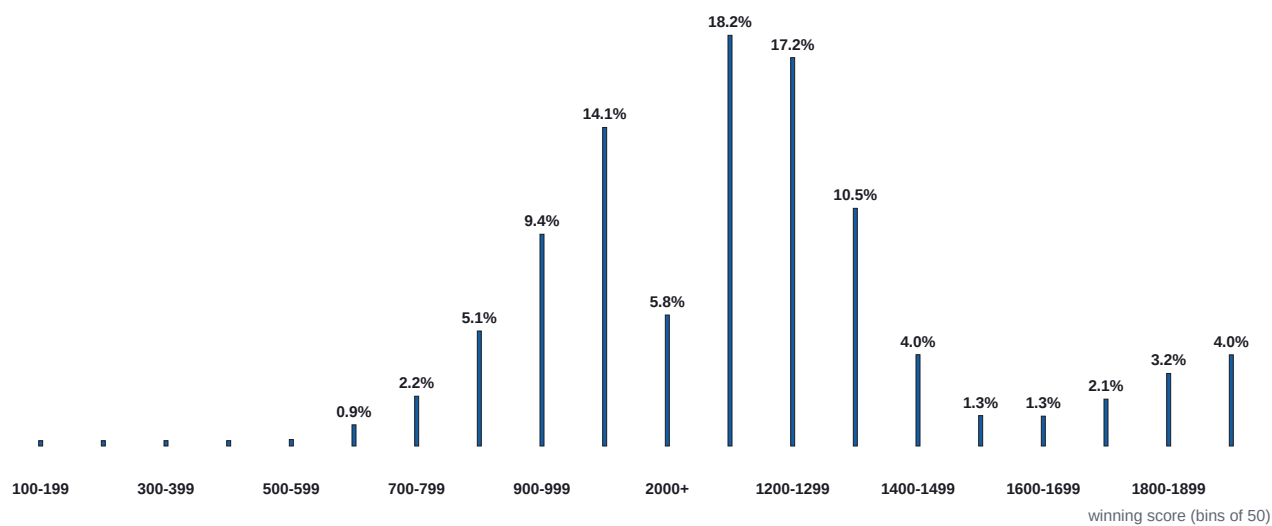


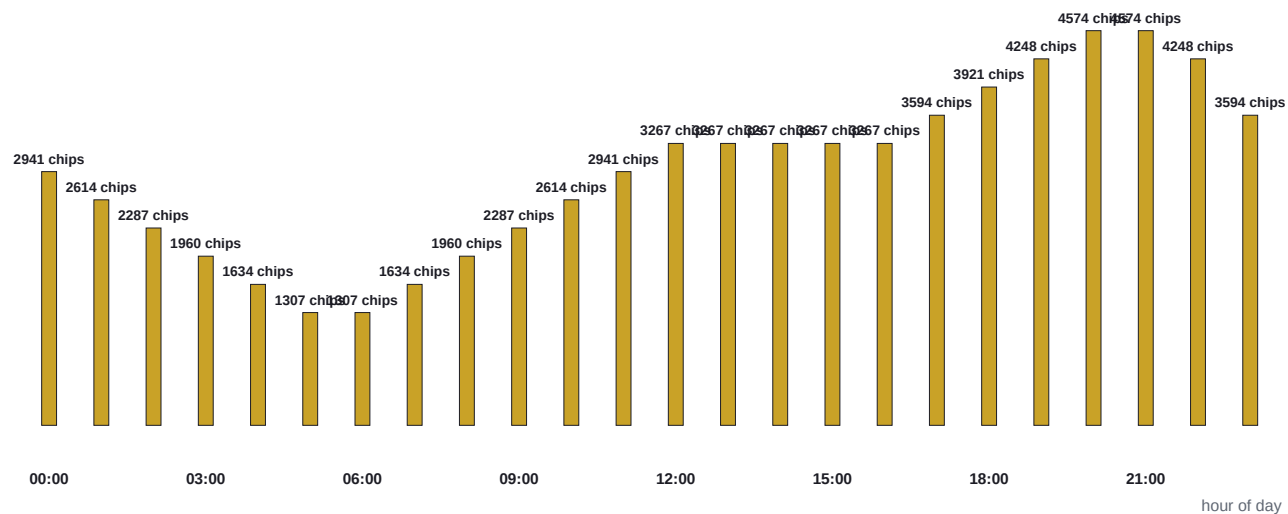
Table point total per round (sum of active seats' final scores) - 2,000,000 rounds



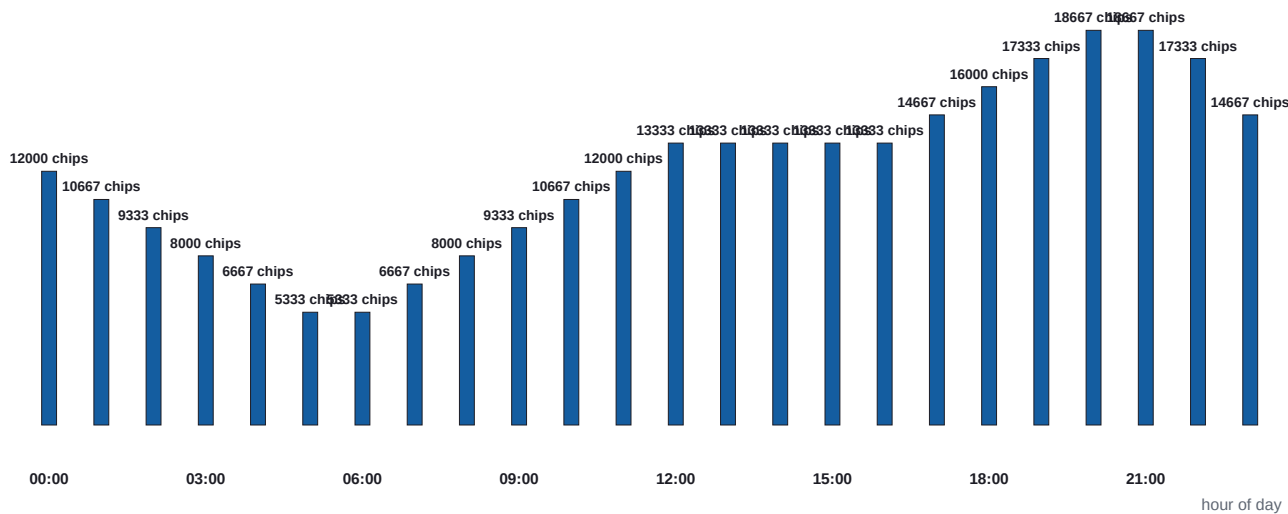
24-hour distribution

The 2,000,000-round audit is equivalent to 3,472.2 standard table-days (576 rounds/day at 24 rounds/hour). The chart below distributes one representative day across 24 hours using a typical casino traffic curve - quiet from 4am to 7am, peaking 8pm to 11pm - and the audit's verified per-round rates.

House win per hour - standard day, 100-chip ante



Handle per hour - standard day, 100-chip ante



| Hour  | Share  | Rounds | Hands | Handle | House win |
|-------|--------|--------|-------|--------|-----------|
| 00:00 | 4.167% | 24     | 120   | 12,000 | 2,941     |
| 01:00 | 3.704% | 21     | 107   | 10,667 | 2,614     |
| 02:00 | 3.241% | 19     | 93    | 9,333  | 2,287     |
| 03:00 | 2.778% | 16     | 80    | 8,000  | 1,960     |

| Hour  | Share  | Rounds | Hands | Handle | House win |
|-------|--------|--------|-------|--------|-----------|
| 04:00 | 2.315% | 13     | 67    | 6,667  | 1,634     |
| 05:00 | 1.852% | 11     | 53    | 5,333  | 1,307     |
| 06:00 | 1.852% | 11     | 53    | 5,333  | 1,307     |
| 07:00 | 2.315% | 13     | 67    | 6,667  | 1,634     |
| 08:00 | 2.778% | 16     | 80    | 8,000  | 1,960     |
| 09:00 | 3.241% | 19     | 93    | 9,333  | 2,287     |
| 10:00 | 3.704% | 21     | 107   | 10,667 | 2,614     |
| 11:00 | 4.167% | 24     | 120   | 12,000 | 2,941     |
| 12:00 | 4.630% | 27     | 133   | 13,333 | 3,267     |
| 13:00 | 4.630% | 27     | 133   | 13,333 | 3,267     |
| 14:00 | 4.630% | 27     | 133   | 13,333 | 3,267     |
| 15:00 | 4.630% | 27     | 133   | 13,333 | 3,267     |
| 16:00 | 4.630% | 27     | 133   | 13,333 | 3,267     |
| 17:00 | 5.093% | 29     | 147   | 14,667 | 3,594     |
| 18:00 | 5.556% | 32     | 160   | 16,000 | 3,921     |
| 19:00 | 6.019% | 35     | 173   | 17,333 | 4,248     |
| 20:00 | 6.481% | 37     | 187   | 18,667 | 4,574     |
| 21:00 | 6.481% | 37     | 187   | 18,667 | 4,574     |
| 22:00 | 6.019% | 35     | 173   | 17,333 | 4,248     |
| 23:00 | 5.093% | 29     | 147   | 14,667 | 3,594     |

Day totals: 576 rounds, 2,880 hands, 288,000 chips handle, 70,576 chips house win (24.506% hold). At \$1 chips that is about \$70.6k of table win per standard day; the peak hour (21:00-22:00) contributes about \$5.0k.

## 14.12 Highly variable 2,000,000-round game day

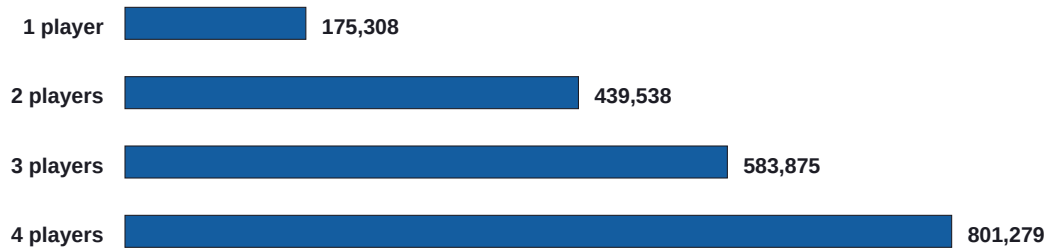
A second 2,000,000-round simulation models an intentionally volatile day: players drop in and out aggressively (20% leave / 40% join probabilities), table sizes swing from heads-up to full, and every ante is drawn uniformly across the full 1-500 chip range instead of the mixed bands used in Section 13.9.

| Day-audit metric     | Value                                                |
|----------------------|------------------------------------------------------|
| Rounds played        | 2,000,000                                            |
| Drop-ins / drop-outs | 800,099 / 800,097                                    |
| Average player seats | 3.006                                                |
| Rounds by table size | 1p: 8.765%   2p: 21.977%   3p: 29.194%   4p: 40.064% |
| Handle per round     | 722.905 chips                                        |
| House win per round  | 176.837 chips                                        |
| Hold percentage      | 24.462%                                              |

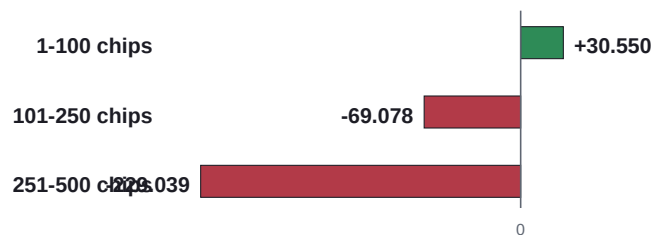
| Day-audit metric          | Value          |
|---------------------------|----------------|
| Player (YOU) EV per round | -129.100 chips |
| Dealer seat EV per round  | +53.851 chips  |

With uniform antes, the average ante is much higher than in the mixed model, so per-round money flows are larger; the hold stays at 24.462%. The dealer posts a fixed 100 chips and wins 31.930% of rounds, making the dealer seat strongly positive in this environment - another argument for the ante spread limit.

### Rounds by table size - highly variable day, 2,000,000 rounds



### YOU EV by ante band - highly variable day

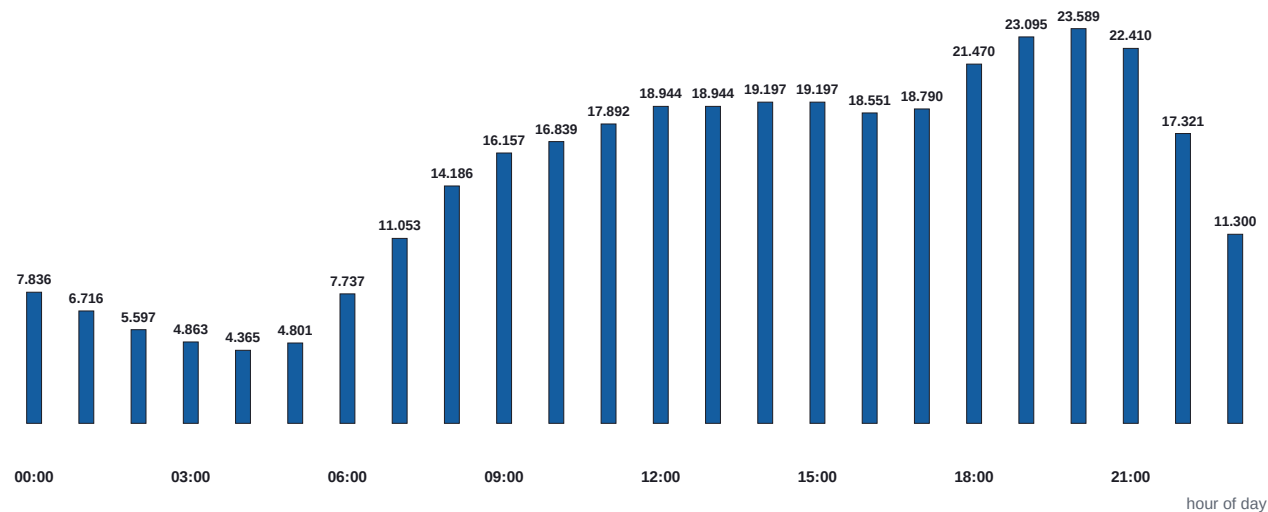




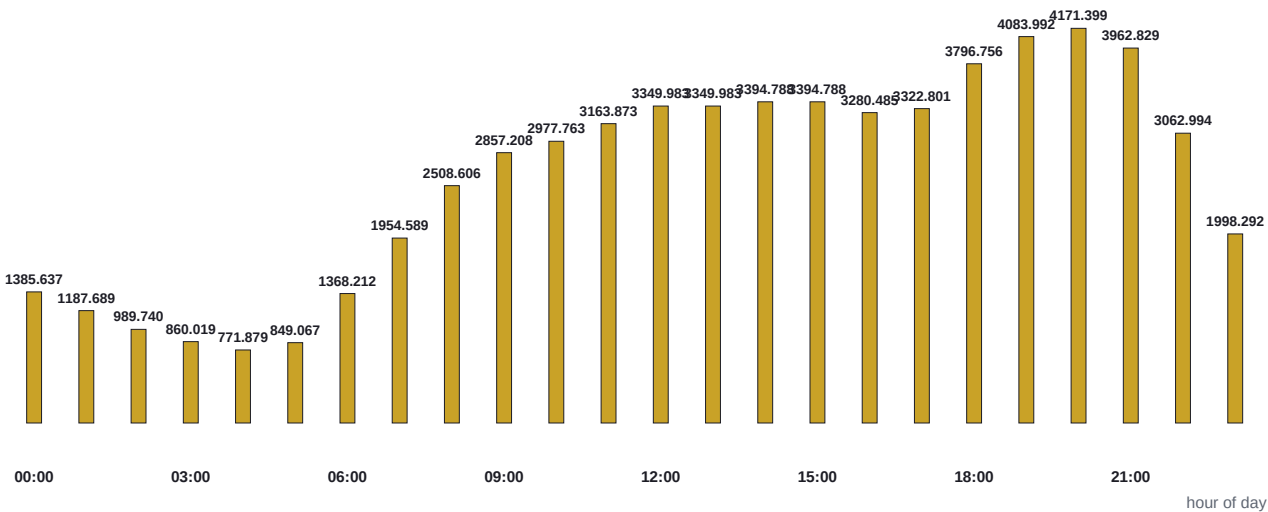
The variable day across 24 hours

The hourly schedule below combines three effects: traffic occupancy (0.200 to 1.000), the player-skill mix per hour (beginners cluster in the afternoon, experienced players in the evening), and the resulting pace from the round-time model (Section 12.8). Per-hour volumes use the 2M day audit's per-round averages.

Rounds per hour - highly variable day (skill mix + occupancy)



House win per hour - highly variable day



| Hour  | Occ.  | Mix B/I/A         | rph    | Rounds | Hands  | Handle   | House win |
|-------|-------|-------------------|--------|--------|--------|----------|-----------|
| 00:00 | 0.350 | 0.300/0.500/0.200 | 22.388 | 7.836  | 23.551 | 5664.456 | 1385.637  |
| 01:00 | 0.300 | 0.300/0.500/0.200 | 22.388 | 6.716  | 20.186 | 4855.248 | 1187.689  |
| 02:00 | 0.250 | 0.300/0.500/0.200 | 22.388 | 5.597  | 16.822 | 4046.040 | 989.740   |

| Hour  | Occ.  | Mix B/I/A         | rph    | Rounds | Hands  | Handle    | House win |
|-------|-------|-------------------|--------|--------|--------|-----------|-----------|
| 03:00 | 0.220 | 0.350/0.450/0.200 | 22.106 | 4.863  | 14.617 | 3515.741  | 860.019   |
| 04:00 | 0.200 | 0.400/0.400/0.200 | 21.825 | 4.365  | 13.119 | 3155.425  | 771.879   |
| 05:00 | 0.220 | 0.400/0.400/0.200 | 21.825 | 4.801  | 14.431 | 3470.968  | 849.067   |
| 06:00 | 0.350 | 0.350/0.450/0.200 | 22.106 | 7.737  | 23.254 | 5593.225  | 1368.212  |
| 07:00 | 0.500 | 0.350/0.450/0.200 | 22.106 | 11.053 | 33.221 | 7990.322  | 1954.589  |
| 08:00 | 0.650 | 0.400/0.400/0.200 | 21.825 | 14.186 | 42.637 | 10255.133 | 2508.606  |
| 09:00 | 0.750 | 0.450/0.350/0.200 | 21.543 | 16.157 | 48.562 | 11680.209 | 2857.208  |
| 10:00 | 0.800 | 0.500/0.350/0.150 | 21.049 | 16.839 | 50.611 | 12173.035 | 2977.763  |
| 11:00 | 0.850 | 0.500/0.350/0.150 | 21.049 | 17.892 | 53.774 | 12933.850 | 3163.873  |
| 12:00 | 0.900 | 0.500/0.350/0.150 | 21.049 | 18.944 | 56.937 | 13694.665 | 3349.983  |
| 13:00 | 0.900 | 0.500/0.350/0.150 | 21.049 | 18.944 | 56.937 | 13694.665 | 3349.983  |
| 14:00 | 0.900 | 0.450/0.400/0.150 | 21.330 | 19.197 | 57.699 | 13877.829 | 3394.788  |
| 15:00 | 0.900 | 0.450/0.400/0.150 | 21.330 | 19.197 | 57.699 | 13877.829 | 3394.788  |
| 16:00 | 0.850 | 0.400/0.400/0.200 | 21.825 | 18.551 | 55.756 | 13410.558 | 3280.485  |
| 17:00 | 0.850 | 0.350/0.450/0.200 | 22.106 | 18.790 | 56.475 | 13583.547 | 3322.801  |
| 18:00 | 0.950 | 0.300/0.450/0.250 | 22.600 | 21.470 | 64.531 | 15521.063 | 3796.756  |
| 19:00 | 1.000 | 0.250/0.450/0.300 | 23.095 | 23.095 | 69.413 | 16695.278 | 4083.992  |
| 20:00 | 1.000 | 0.200/0.450/0.350 | 23.589 | 23.589 | 70.898 | 17052.596 | 4171.399  |
| 21:00 | 0.950 | 0.200/0.450/0.350 | 23.589 | 22.410 | 67.353 | 16199.966 | 3962.829  |
| 22:00 | 0.750 | 0.250/0.450/0.300 | 23.095 | 17.321 | 52.059 | 12521.459 | 3062.994  |
| 23:00 | 0.500 | 0.300/0.450/0.250 | 22.600 | 11.300 | 33.964 | 8168.980  | 1998.292  |

Highly variable day totals: 350.851 rounds, 1054.506 hands, 253632.084 chips handle, 62043.370 chips house win (24.462% hold). The peak evening hours carry roughly double the morning volume.

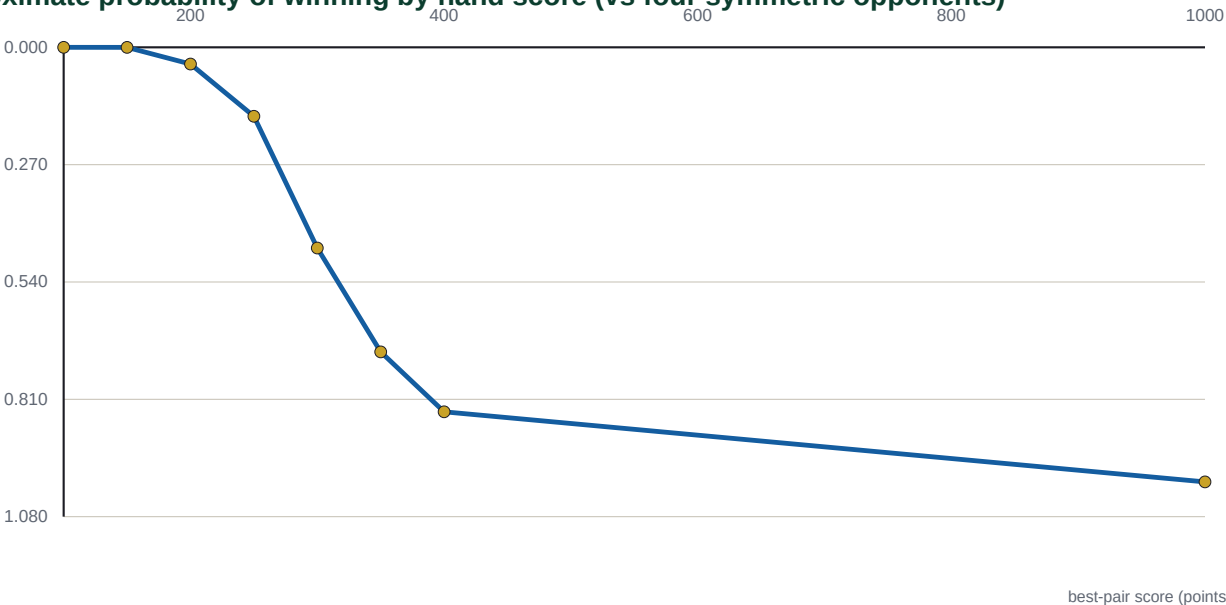
14.13 Player odds display & verification

Players must be able to see and verify the odds at the table without trusting the house. Spit Shine publishes a fixed odds card, shows live probabilities in the digital client, and makes every figure reproducible from the audit datasets.

The published odds card (three decimal places)

| Odds statement                                | Value   |
|-----------------------------------------------|---------|
| House edge (hold) - competitive full table    | 24.503% |
| Break-even win rate for a player (no Bow Out) | 26.667% |
| Break-even win rate to stay vs Bow Out        | 6.667%  |
| P(Double Aces in your 4 cards)                | 2.895%  |
| P(someone dealt Double Aces per round)        | 13.917% |
| P(win with a 200-point hand, approximate)     | 3.847%  |
| P(win with a 300-point hand, approximate)     | 46.183% |
| P(win with a 400-point hand, approximate)     | 83.871% |
| Exact-tie (pot split) frequency               | 3.165%  |

Approximate probability of winning by hand score (vs four symmetric opponents)



How the odds are verified: (1) the canonical engine is seeded and deterministic, so any figure can be reproduced; (2) every round is audit-logged with card-level detail and exported as JSON/CSV; (3) the digital client shows live win probability and hand tier during play; (4) an independent laboratory (GLI-class) certifies the published payout tables; (5) daily RNG statistical tests confirm shuffle randomness.

Balance policy: the house edge is set by published rules (25% rake, dealer two-card Shine, tie-break order) and is never changed mid-session. The cooperative attraction package is the only intentional player-positive offering, and it is capped, advertised as an event promotion, and justified by conversion economics (Section 14.5).

# 15. Integrity, Fairness & Regulatory Readiness

## 15.1 Randomness

The reference implementation uses a seeded Mersenne Twister for reproducibility and testing. A live casino deployment requires a certified hardware RNG (NIST SP 800-90A, FIPS 140-2/3 validated) with continuous statistical testing (NIST STS, Dieharder) and documented seeding for every shuffle.

## 15.2 Audit trail

Every round is recorded with card-level detail: deal, showing, Bow Out refunds, Spit swaps (who gave what to whom), Shine discards and draws, pair selections, showdown scores, tie-breaks, rake, and payouts. The log is exported as JSON and CSV from the client (AUDIT button) and is the basis for dispute resolution and regulator reporting.

## 15.3 Regulatory readiness checklist

| Requirement                                        | Status                                 | Owner       |
|----------------------------------------------------|----------------------------------------|-------------|
| Single canonical rules engine                      | In place (v11)                         | Development |
| Certified hardware RNG + daily statistical tests   | Open                                   | Vendor / IT |
| Card-level audit log with integrity (HMAC/keyed)   | Log in place; keying open              | Development |
| Shuffle / penetration / burn procedures            | Documented in this guide               | Operations  |
| Independent game-math certification (e.g., GLI)    | Open                                   | Testing lab |
| Table limits, dispute and surveillance procedures  | Open                                   | Operations  |
| Spit order rotation for positional fairness        | Recommended - implement in floor rules | Operations  |
| AML / KYC / responsible-gaming integration         | Open                                   | Compliance  |
| Cooperative mode economics rebalanced or withdrawn | Open                                   | Product     |

## 16. Edge Cases, Official Rulings & Ambiguities

This section collects every corner case identified during the audit and assigns an official ruling. Where the earlier prototypes or the design documents disagreed, the ruling below is the canonical answer; where a question cannot be settled without jurisdiction- or venue-specific policy, it is marked OPEN and must be decided before certification.

### 16.1 Table and seating edge cases

| Situation                           | Ruling                                                                                                                                     |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Player joins during a round         | Not allowed. Seats change only between rounds; the new player posts an ante and receives a fresh hand next round.                          |
| Player leaves mid-round             | Not allowed for the current round; the hand plays out and the seat is removed at round end.                                                |
| All players leave the table         | The table closes; the dealer seat is not dealt for an empty table.                                                                         |
| Fewer than four players             | Fully supported: empty seats post no ante and take no cards; the pot and rake scale with occupied seats.                                   |
| A seat posts an ante outside limits | The dealer rejects the ante; the seat must adjust to the published minimum/maximum before cards are dealt.                                 |
| Mixed antes at one table            | Allowed, subject to the recommended 5x spread limit (Section 14.9).                                                                        |
| Shoe used for multiple rounds       | Live floors may run a continuous shoe per Section 7; the canonical engine reshuffles every round. Per-round odds are identical either way. |

### 16.2 Deal and card-handling edge cases

| Situation                                    | Ruling                                                                                                                               |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Card exposed during the deal                 | Dead card: place in the discard rack and draw a replacement for that seat.                                                           |
| Seat dealt the wrong number of cards         | Misdeal: void the round, return all antes, and redeal from the shoe (reshuffle if the cut card appears).                             |
| Duplicate cards (two decks)                  | Legal and expected; exact duplicates are distinct physical cards and need no special rule.                                           |
| Burn card                                    | One card is burned face-down at the start of each new shoe; it plays no part in any round.                                           |
| Cut card appears mid-round                   | Finish the round, then reshuffle before the next deal.                                                                               |
| Shoe exhausted mid-round                     | Impossible within the canonical limits (max 26 cards per round vs 104 in the shoe); if it ever occurs, void the round and reshuffle. |
| Player exposes a face-down card accidentally | The card stays face-up (it becomes known information) and the round continues; penalties, if any, follow house policy.               |

### 16.3 Phase-level edge cases

| Phase       | Situation                                  | Ruling                                                        |
|-------------|--------------------------------------------|---------------------------------------------------------------|
| 2 - Showing | Player changes their mind after confirming | Confirmation is final; shown cards stay face-up for the round |

| Phase         | Situation                               | Ruling                                                                                   |
|---------------|-----------------------------------------|------------------------------------------------------------------------------------------|
| 2 - Showing   | Player shows all four cards             | Allowed; Bow Out is locked and the seat cannot Spit or Shine (no face-down cards remain) |
| 3 - Bow Out   | Player with shown cards presses Bow Out | Rejected - showing locks Bow Out; the hand plays out                                     |
| 3 - Bow Out   | Refund amount for a non-standard ante   | 25% of that seat's ante, floored to the nearest chip                                     |
| 4 - Spit      | Actor has no face-down cards            | Cannot Spit; the seat passes                                                             |
| 4 - Spit      | Target has no face-down cards           | Target is not selectable; choose another seat                                            |
| 4 - Spit      | Target already spat at this round       | Not selectable - each target receives at most one Spit                                   |
| 4 - Spit      | Human Spits at the dealer               | Allowed; the dealer can receive a Spit but never initiates one                           |
| 4 - Spit      | Received card visibility                | Private to the spitter only; never table-visible                                         |
| 5 - Shine     | Player has no face-down cards           | Cannot Shine; the seat keeps its hand                                                    |
| 5 - Shine     | Player discards without drawing         | Never allowed - a replacement is always drawn from the shoe                              |
| 6 - Selection | Human picks a non-optimal pair          | Allowed - human choice is final; AI always optimizes                                     |
| 6 - Selection | Bowed-out seat selection                | None required; bowed seats are excluded from showdown                                    |

## 16.4 Showdown and payout edge cases

| Situation                              | Ruling                                              | Worked example                                                                                |
|----------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------|
| All players Bow Out (competitive)      | The dealer is the only active seat and wins         | 4 players bow: pot 400, rake 100, dealer receives 300 (net +200); each player -75; house +100 |
| All players Bow Out (cooperative)      | The dealer wins; the house keeps the entire pot     | Pot 400, house +400; each player -100                                                         |
| Three-way exact tie                    | Pot splits three ways; remainder to house           | Pot 500: payout 375 = 125 each                                                                |
| Two-way tie on an odd payout           | Floor split; remainder to house as rounding rake    | Pot 475: payout 357 = 178 + 178 + 1                                                           |
| Two players both hold Double Aces      | Apply score, then rank, then suit; exact tie splits | A(spades)+A(hearts) beats A(diamonds)+A(clubs) on suit; identical suit-max splits             |
| Cooperative tie (player equals dealer) | Dealer wins ties by rule                            | Player 300 vs dealer 300: dealer wins; house keeps pot                                        |
| Payout dispute                         | The audit log is authoritative                      | Round events include every card, action, score, and payout                                    |

## 16.5 Digital and POC behavior

| Situation                         | Ruling                                                                                                                       |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Empty seats in the digital client | AI fills seats by default for the 4-player demo; the lobby can remove AI seats for heads-up play                             |
| Player disconnects mid-round      | 60-second grace period; after that the AI takes over the seat for the remainder of the round; the seat may rejoin next round |
| Bankroll protection on disconnect | The round settles normally; the seat's result is credited when the player reconnects                                         |
| Audit access                      | Every round is exported to JSON/CSV via the in-game AUDIT button                                                             |

## 16.6 Official rulings quick reference

| Question                                        | Answer                                                                          |
|-------------------------------------------------|---------------------------------------------------------------------------------|
| Who wins an exact tie?                          | The pot splits among tied seats.                                                |
| Can the dealer Bow Out or Spit?                 | No - never.                                                                     |
| Is the Spit-received card public?               | No - private to the spitter.                                                    |
| Can a shown card be hidden again?               | No - showing is final for the round.                                            |
| What is the minimum/maximum ante?               | 1 chip / 500 chips.                                                             |
| How many players can sit?                       | 1-4 plus the dealer; capacity 4 player seats.                                   |
| When can players join or leave?                 | Between rounds only.                                                            |
| What happens if the cut card appears mid-round? | Finish the round, then reshuffle.                                               |
| Who handles a payout dispute?                   | The floor, using the audit log.                                                 |
| Is the house rake always 25%?                   | Yes, of the pot after Bow Out refunds; rounding remainder stays with the house. |

## 16.7 Open questions (decision required before certification)

- Jurisdictional table limits and currency conversion policy (OPEN - venue/regulator).
- Cooperative mode rebalance package approval (proposed in Section 14.5; pending formal certification).
- Side wagers and tournament variants (not part of the current rules; each requires independent math certification).
- Ante spread limit value (5x recommended; final value is a commercial decision).



## 17. Known Issues & Remediation

The audit identified defects across the earlier prototypes. All gameplay-critical defects are fixed in the canonical engine v11; the register below is retained for transparency and for any laboratory reviewing the development history.

| ID  | Severity | Issue                                                                          | Status                        |
|-----|----------|--------------------------------------------------------------------------------|-------------------------------|
| B1  | Critical | v7 GUI human Spit targeted self; v7 sim Spit was a no-op                       | Fixed in v11                  |
| B2  | Critical | v9/v10 editions imported an unavailable copperhead module and could not launch | Superseded by v12 client      |
| B3  | High     | Spat-at-once rule tracked the wrong set - multiple Spits per target possible   | Fixed in v11                  |
| B4  | High     | Fixed Spit order created seat-position bias                                    | Rotation recommended          |
| B5  | High     | AI Bow Out missing in interactive v7/v10                                       | Fixed in v11                  |
| B6  | High     | White paper odds/EV errors (Double Aces, house edge, dealer EV)                | Corrected in this guide       |
| B7  | Medium   | Exact ties awarded to earliest seat instead of split                           | Fixed in v11                  |
| B8  | Medium   | Cooperative all-Bow-Out crash                                                  | Fixed in v11                  |
| B9  | Medium   | Cooperative house edge ~46% undisclosed                                        | Flagged for rebalance         |
| B10 | Medium   | Dealer Spit inconsistent between v4 and v7/v9/v10                              | Canonical: dealer never Spits |
| B11 | Low      | Tie-break suit semantics inconsistent (max vs sum)                             | Fixed in v11                  |
| B12 | Low      | Integer split remainder unaccounted                                            | Defined as rounding rake      |

## 18. Appendix A - Glossary

| Term                | Definition                                                                                 |
|---------------------|--------------------------------------------------------------------------------------------|
| Ante                | Fixed wager placed by each seat at the start of a round (default 100).                     |
| Bow Out             | Phase 3 option to forfeit and recover 25% of the ante.                                     |
| Dealer              | Seat 4; never Bows Out, never Spits, may Shine 0-2 cards.                                  |
| Double Aces         | A+A pair worth 1000 points, the highest hand.                                              |
| Face-down / Face-up | Hidden versus table-visible card state.                                                    |
| House rake          | 25% of the pot taken at showdown.                                                          |
| Pot                 | Sum of antes minus Bow Out refunds.                                                        |
| Shine               | Phase 5 discard-and-redraw; players 0-1, dealer 0-2.                                       |
| Showdown            | Phase 7 comparison: score, high rank, high suit, split.                                    |
| Spit                | Phase 4 exchange: give one face-down card, take one; received card private to the spitter. |
| Shoe                | The 104-card dealing container.                                                            |
| Split               | Exact-tie resolution that divides the pot equally.                                         |

## 19. Appendix B - Worked Example: One Full Round

The round below is a fully scripted competitive round that exercises every mechanic: Showing, Bow Out, Spit, Shine, tie-breaking, rake, and payouts. Cards are fixed for illustration; in live play every card is dealt from the shoe.

### B.1 Initial deal

| Seat   | Cards dealt                               | Best pair  | Tier     |
|--------|-------------------------------------------|------------|----------|
| YOU    | K(spades) Q(diamonds) 7(clubs) 3(hearts)  | K+Q = 350  | Strong   |
| ALICE  | A(hearts) 9(spades) 8(diamonds) 2(clubs)  | A+9 = 190  | Marginal |
| BOB    | J(hearts) 7(diamonds) 5(clubs) 4(spades)  | J+7 = 170  | Weak     |
| CAROL  | 10(spades) 6(hearts) 5(diamonds) 3(clubs) | 10+6 = 160 | Weak     |
| DEALER | K(clubs) 10(hearts) 8(diamonds) 3(spades) | K+10 = 300 | Solid    |

### B.2 Phase decisions

| Phase       | Decision                                                           | Why                                                                                                                                                     |
|-------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 - Showing | YOU shows K(spades); ALICE, BOB, CAROL hide; DEALER shows K(clubs) | YOU and the dealer have 350/300 hands and show to deter Spits; the marginal hands hide to keep Bow Out                                                  |
| 3 - Bow Out | BOB Bows Out (refund 25); ALICE and CAROL stay                     | BOB's 170-point hand is below the fold line; ALICE and CAROL stay despite weak hands                                                                    |
| 5 - Shine   | CAROL discards 2(clubs), draws 9(clubs); everyone else keeps       | CAROL's hand is 180 after her Spit; discarding the 2 and drawing 9(clubs) lifts her best pair to 10+9 = 190. YOU/ALICE/dealer hold 350/190/300 and keep |

### B.3 Spit swaps (Phase 4)

| Actor | Target | Card given | Card taken  | Hand after swap                     |
|-------|--------|------------|-------------|-------------------------------------|
| YOU   | ALICE  | 3(hearts)  | 8(diamonds) | K Q 7 8 - best pair still 350       |
| ALICE | CAROL  | 2(clubs)   | 5(diamonds) | A 9 3 5 - best pair still 190       |
| CAROL | DEALER | 3(clubs)   | 8(diamonds) | 10 6 2 8 - best pair 180 (improved) |

Each target receives exactly one Spit: ALICE from YOU, CAROL from ALICE, DEALER from CAROL. BOB is out of the round and cannot be targeted. The dealer never Spits.

### B.4 Showdown and payouts

| Rank | Seat | Pair  | Score | Notes                                            |
|------|------|-------|-------|--------------------------------------------------|
| 1st  | YOU  | K + Q | 350   | Winner - pot 475, rake 118, payout 357, net +257 |

| Rank | Seat   | Pair   | Score | Notes                                   |
|------|--------|--------|-------|-----------------------------------------|
| 2nd  | DEALER | K + 10 | 300   | Loses 100                               |
| 3rd  | ALICE  | A + 9  | 190   | Beats CAROL on high rank (Ace 14 vs 10) |
| 4th  | CAROL  | 10 + 9 | 190   | Tie with ALICE on score, loses on rank  |
| Out  | BOB    | -      | -     | Bow Out: net -75                        |

Cash-flow check: +257 (YOU) - 100 (ALICE) - 100 (CAROL) - 100 (dealer seat) - 75 (BOB) + 118 (house rake) = 0. Every chip is accounted for, and the same check is performed automatically by the audit log after every round.

# 20. Appendix C - Data Provenance & Reproduction

| Artifact                  | Path / command                                 |
|---------------------------|------------------------------------------------|
| Canonical engine          | spitshine_engine.py                            |
| Audit harness             | spitshine_analysis.py --rounds 20000           |
| Raw audit data            | spitshine_odds_report.json                     |
| Headless integration test | test_aaa_flow.py                               |
| Playable client + Academy | spitshine_aaa.py                               |
| Full audit report         | SPIT_SHINE_AUDIT_REPORT.md (SS-AUDIT-2026-001) |

All Monte Carlo figures use the fixed seed 20260804 (competitive baseline and variants) with 20,000 rounds per scenario, and the canonical threshold policies at every seat. Exact probabilities are computed by enumeration of all  $C(104,4)$  hands. Chip conservation holds to machine precision in every run.

End of document. SS-GUIDE-2026-001 v1.0 - approved for distribution to casino business development and regulatory review.